

WJEC GCE Physics

A2 UNIT 3 PAST PAPER BY TOPIC

IGC HK Exam



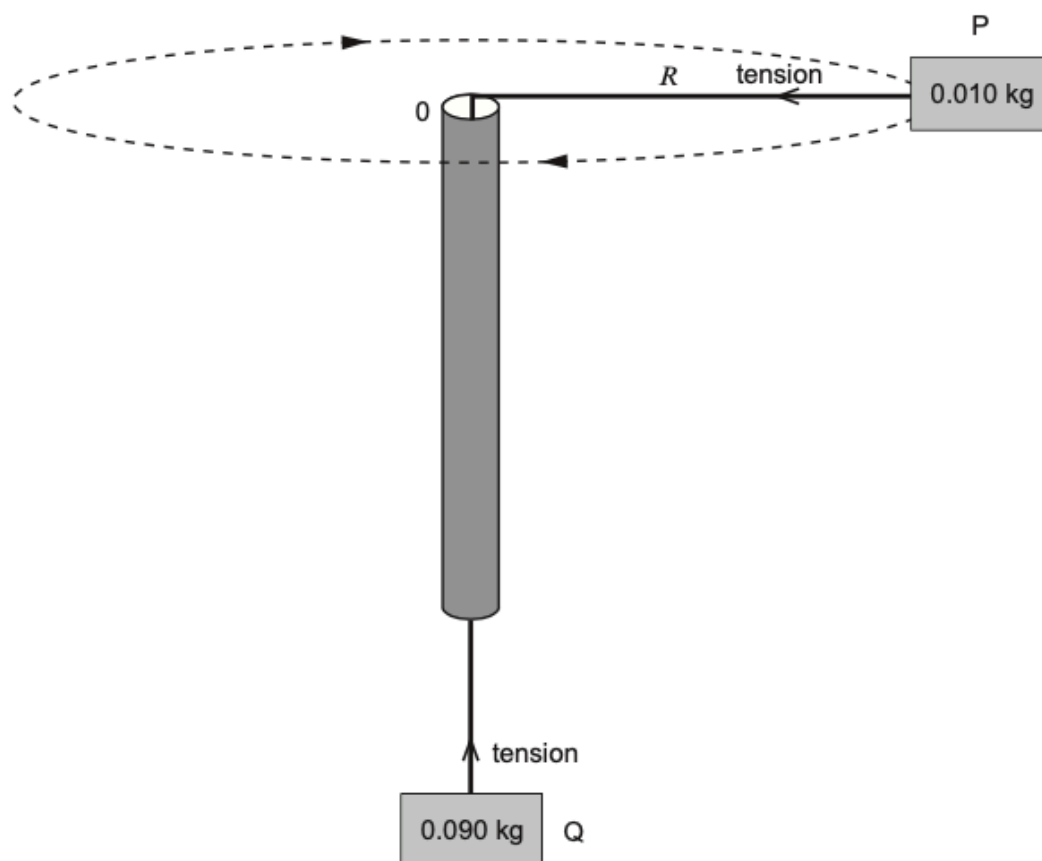
Circular Motion

2017

5. A piece of string is threaded through a hollow narrow cylinder. Two small objects, P and Q, with masses 0.010 kg and 0.090 kg respectively are attached to the ends of the string, as shown.

A student holds the cylinder and sets the 0.010 kg mass rotating in a horizontal circle of radius R , which is kept constant at 0.50 m. The time for 10 rotations is recorded. The tension in the string provides both the centripetal force on P and an upward force to hold Q in equilibrium.

The measurement is repeated for different values of R . All measurements are recorded in the table overleaf.



- (a) Show that the speed, v , of mass P for each measurement is given by:

$$v = \frac{2\pi R}{T} \text{ where } T \text{ is the period of rotation.}$$

[1]

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(b) Complete the table.

[4]

R / m	Time for 10 rotations / s	Period T / s	v / ms^{-1}	$v^2 / \text{m}^2 \text{s}^{-2}$
0.50	4.7			
0.60	5.2			
0.70	5.6			
0.80	6.0			
0.90	6.3			

(c) (i) Assuming that OP is horizontal, write an equation relating the centripetal force to v and R . [2]

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(ii) Hence, by using the equation for the forces acting on mass Q, show that:

$$v^2 = 9g R$$

where g is the acceleration due to gravity.

[3]

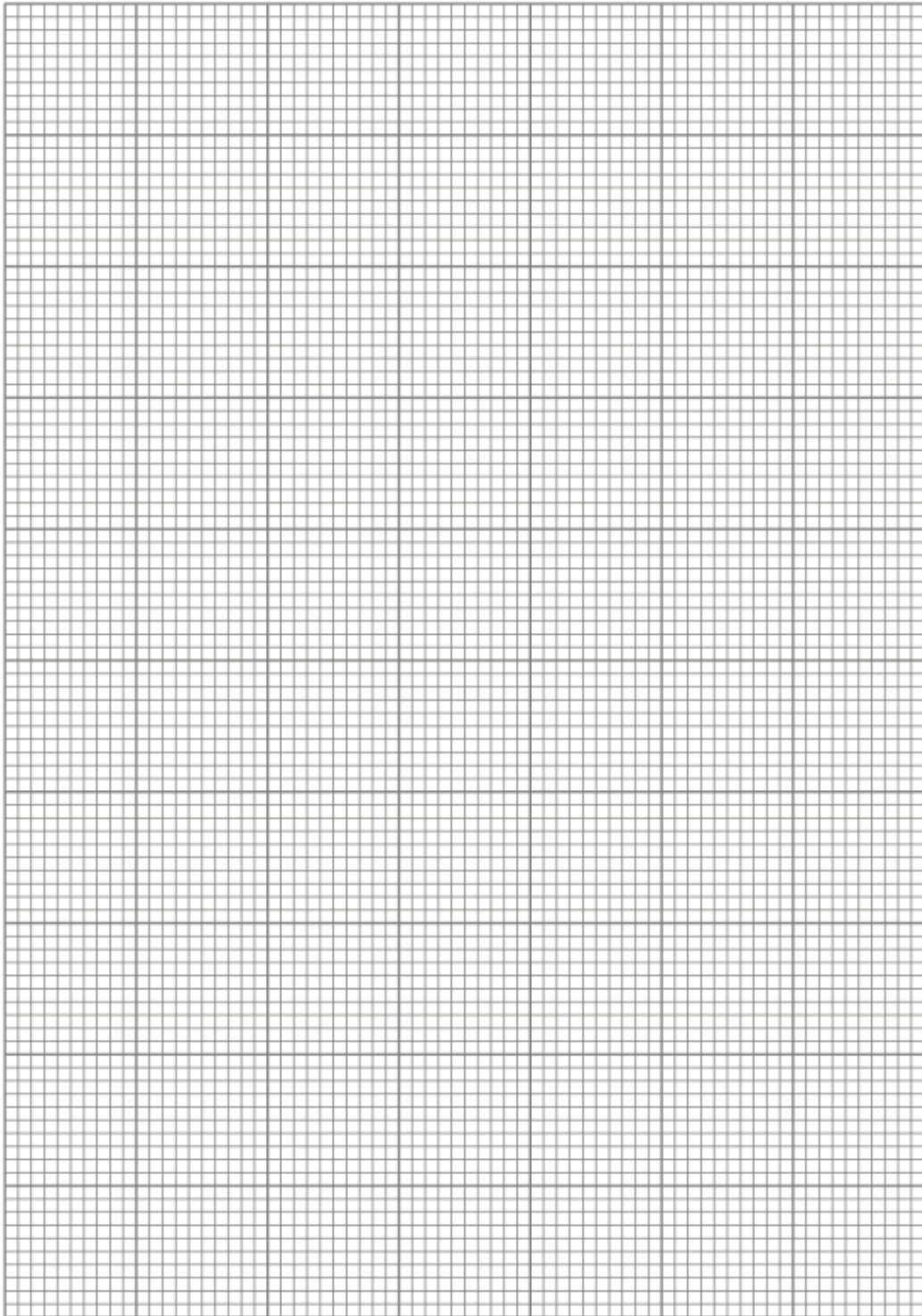
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- (d) (i) Use the data in the table to plot a graph of v^2 (y -axis) against R (x -axis). [4]



(ii) Determine a value for g .

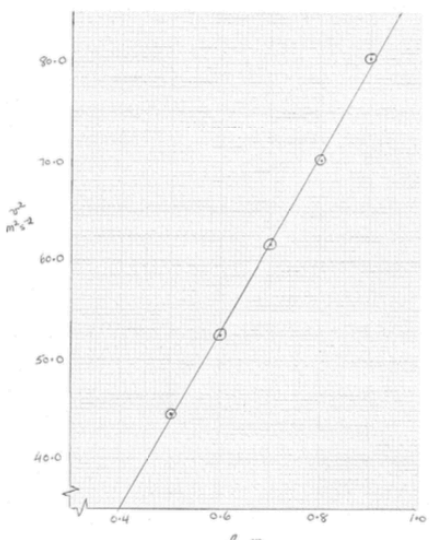
[3]

(iii) Suggest a way in which the experiment can be improved.

[1]

Question		Marking details	Marks available				Maths	Prac																														
			AO1	AO2	AO3	Total																																
5	(a)	Distance moved by mass P in 1 period, T (i.e. in one rotation) = $2\pi R$ and speed = distance / time = $\frac{2\pi R}{T}$ Alternative: $\omega = \frac{2\pi}{T}$ and $v = \omega R$		1		1	1	1																														
	(b)	<table border="1"><thead><tr><th>R / m</th><th>Time of 10 rot / s</th><th>T / s</th><th>$v / m s^{-1}$</th><th>$v^2 / m^2 s^{-2}$</th></tr></thead><tbody><tr><td>0.50</td><td>4.7</td><td>0.47</td><td>6.68</td><td>44.6</td></tr><tr><td>0.60</td><td>5.2</td><td>0.52</td><td>7.25</td><td>52.6</td></tr><tr><td>0.70</td><td>5.6</td><td>0.56</td><td>7.85</td><td>61.6</td></tr><tr><td>0.80</td><td>6.0</td><td>0.60</td><td>8.38</td><td>70.2</td></tr><tr><td>0.90</td><td>6.3</td><td>0.63</td><td>8.98</td><td>80.6</td></tr></tbody></table> T column: All values correct (1) For column v all values correct (1) For column v^2 all values correct ecf (1) Consistent use of sig figs in each column and 2 or 3 sig figs (1)	R / m	Time of 10 rot / s	T / s	$v / m s^{-1}$	$v^2 / m^2 s^{-2}$	0.50	4.7	0.47	6.68	44.6	0.60	5.2	0.52	7.25	52.6	0.70	5.6	0.56	7.85	61.6	0.80	6.0	0.60	8.38	70.2	0.90	6.3	0.63	8.98	80.6		4		4	4	4
R / m	Time of 10 rot / s	T / s	$v / m s^{-1}$	$v^2 / m^2 s^{-2}$																																		
0.50	4.7	0.47	6.68	44.6																																		
0.60	5.2	0.52	7.25	52.6																																		
0.70	5.6	0.56	7.85	61.6																																		
0.80	6.0	0.60	8.38	70.2																																		
0.90	6.3	0.63	8.98	80.6																																		

Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
	(c) (i)	Centripetal force = $0.010 \frac{v^2}{R}$ (1 for $\frac{mv^2}{R}$; 1 if value inserted for m)	1	1		2	1	2
	(ii)	Forces acting on mass Q: $0.090g - \tau = 0$ τ : tension (1) So $\tau = 0.090g$. Substitution for τ into (c)(i) (1) $0.090g = 0.010 \frac{v^2}{R}$ $v^2 = \frac{0.090g}{0.010} R$ $v^2 = 9g R$ clear and convincing working (1)		3		3	2	3

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
(d)	(i)		 Axes, suitable choice scales (no multiples of 3) and labels on both axes (1) – scales to occupy more than half of paper All points plotted correctly to $\pm\frac{1}{2}$ small square division (2) 4 points plotted correctly to $\pm\frac{1}{2}$ small square division award 1 mark 1-3 points plotted correctly to $\pm\frac{1}{2}$ small square division award 0 marks Line of best fit (1)		4		4	4	4

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(ii)		gradient = $\frac{50.0}{0.56} = 89.286 \text{ m s}^{-2}$; find gradient from best fit line (1) also gradient = $9g$ general method (1) $9g = 89.286$ $g = \frac{1}{9} 89.286 = 9.92 \text{ m s}^{-2}$ unit mark (1) (Accept $g = 8.8$ to 10.8 m s^{-2} i.e. uncertainty of ~10%.) Use of single data point award a maximum of 2 marks			3	3	3	3
	(iii)		Take measurements for each value of R several times or measure time of more rotations or use of video capture or increase radius and period Accept repeat readings Don't accept have an assistant			1	1		1
			Question 5 total	1	13	4	18	15	18

2018

1. (a) A car travels at a constant speed of 45.0 km h^{-1} around a curve in the road with a radius of 80 m.

(i) Explain why the car is accelerating. [2]

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(ii) Calculate the angular velocity of the car (in rad s^{-1}) as it travels around the curve. [2]

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(iii) Calculate the acceleration of the car and state its direction. [3]

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(b) Discuss how the application of science enables cars to travel safely around curves. [2]

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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
1	(a)	(i)	Velocity [of car] is changing [with time] (1) because its direction is changing [so the car is accelerating] (1) or There is a <u>resultant</u> force (1) [towards the centre] due to friction / grip (1)	2			2		
		(ii)	$v = \frac{45 \times 10^3}{60 \times 60} = 12.5 \text{ m s}^{-1}$ (conversion) (1) $\omega = \frac{v}{r} = \frac{12.5}{80} = 0.156 \text{ rad s}^{-1}$ substitution and calculation (1)		2		2	2	
		(iii)	$a = \frac{v^2}{r} = \frac{(12.5)^2}{80}$ substitution (ecf) (1) [Alt: use $a = \omega^2 r$] $= 1.95 \text{ m s}^{-2}$ (1) Direction: towards centre (of 'circular' motion) (1)	1 1	1		3	2	
	(b)		Either: Any two $\times$ (1) of these points - Appropriate tyre design for friction - Banking of road [for contribution from normal contact force] - Appropriate surface - Suspension set-up - Anti-roll bars. (or any sensible answers, one referring to road and the other to the car) Or any one sensible point (1) + explanation of the role of physics (1) [			2	2		
			Question 1 total	4	3	2	9	4	0

2022

4. A satellite orbiting the Earth completes one revolution in 105 minutes.

Mass of the Earth = 6.0×10^{24} kg, radius of the Earth = 6400 km

- (a) Explain what is meant by the term *radian*. [2]

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- (b) Calculate the satellite's angular velocity, ω , in rad s^{-1} . [3]

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- (c) The gravitational force on the satellite is given by $\frac{GM_{\text{E}}m}{R^2}$ where m is the mass of the satellite, M_{E} is the mass of the Earth and R is the radius of the orbit. Show that:

$$R = \sqrt[3]{\frac{GM_{\text{E}}}{\omega^2}} \quad [2]$$

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(d) Determine the height of the satellite above the surface of the Earth. [3]

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(e) The mass of the Moon is 7.3×10^{22} kg and its radius is 1 740 km. Discuss whether a satellite would be able to orbit the Moon with a period of 105 minutes. [3]

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Angle where (1) accept diagram showing angle arc length equals the radius or approximately 57.3° or $2\pi = 360^\circ$ (1)	2			2		
	(b)		Period, $T = 105 \times 60 = 6300$ [s] or 105 [min] (1) Substitution (regardless of unit): $\omega = \frac{2\pi}{T} = \frac{2\pi}{6300}$ (1) $= 9.97 \times 10^{-4}$ [rad s ⁻¹](1)	1	1 1		3	3	
	(c)		$m\omega^2 R = \frac{GM_E m}{R^2}$ (1) Accept $\frac{mv^2}{r} = \frac{GM_E m}{R^2}$ $R^3 = \frac{GM_E}{\omega^2}$ [so: $R = \sqrt[3]{\frac{GM_E}{\omega^2}}$] (1)		2		2	1	
	(d)		Substitution: $R = \sqrt[3]{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(9.97 \times 10^{-4})^2}}$ ecf (1) $R = 7\,380$ k[m] (1) Altitude above the surface of the Earth = $7\,380 - 6\,400 = 980$ k[m] (1)	1	1 1		3	3	
	(e)		$R = \sqrt[3]{\frac{(6.67 \times 10^{-11})(7.3 \times 10^{22})}{(9.97 \times 10^{-4})^2}}$ ecf (1) $R = 1\,698$ k[m] (1) Correct conclusion based on candidate's answer e.g. not possible since inside Moon (1) Alternative: Substituting into equation to calculate period or to calculate mass (1) Correct answer e.g. to period (1) Correct conclusion based on period / mass e.g. period longer so not possible, mass smaller so not possible (1)			3	3	1	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Alternative: Can also work centripetal acceleration (1.7) (1) Calculate gravitational acceleration (1.6) (1) Not possible (since gravity too weak) (1)						
			Question 4 total	4	6	3	13	8	0

2023

2. (a) Explain what is meant by centripetal force and state its direction. [2]

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- (b) A small sphere of mass 30 g at the end of a light string rotates in a horizontal circle of radius 0.80 m and completes 10 revolutions in 15 s.

- (i) Show that the speed of the sphere is 3.35 m s^{-1} . [2]

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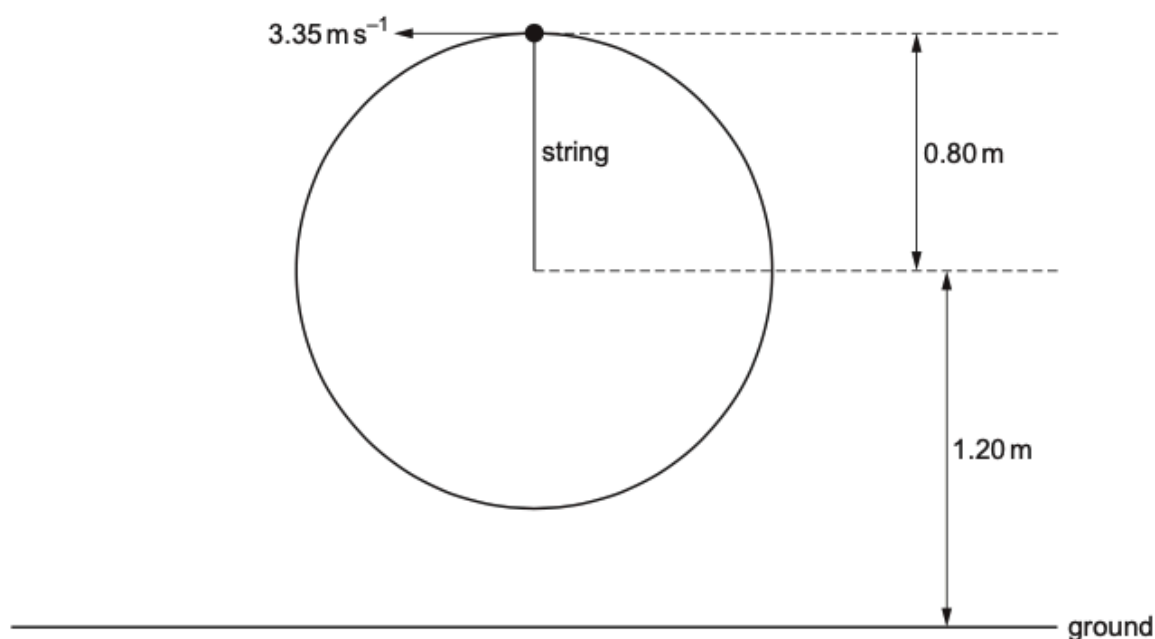
- (ii) Calculate the centripetal force on the sphere. [2]

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- (c) The sphere now rotates in a **vertical** circle of the same radius, from a point that is 1.20 m above the ground. The speed at the top of the circle is 3.35 m s^{-1} .



Calculate the tension in the string at the **top of the circle**.

[2]

- (d) (i) Calculate the speed of the sphere when it reaches the **bottom of the circle**. [3]

- (ii) A student claims that if the string breaks when the sphere is at the **top of the circle** it will reach the ground at a horizontal distance of approximately 2m away from the point of release. Investigate if her claim is correct, justifying your answer.

[4]

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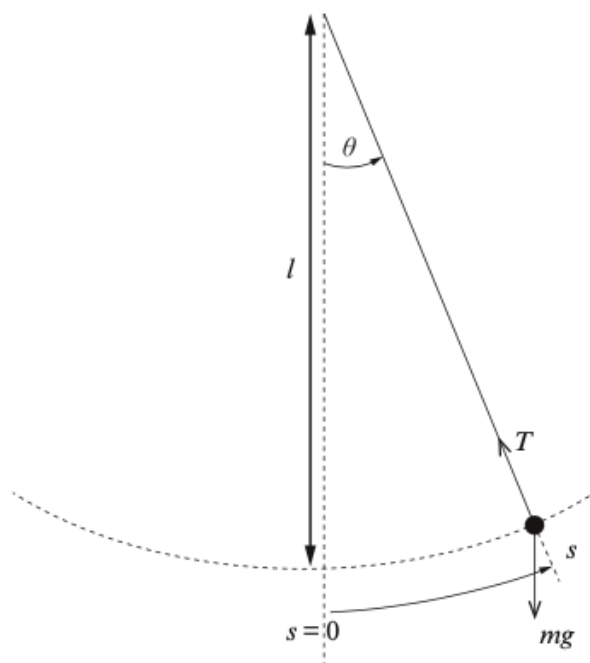
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Is the [resultant] force for circular motion (1) It is directed towards the centre [of the circle] (1)	2			2		
	(b)	(i)		$T = \frac{15}{10} = 1.5 \text{ s}$ or $f = \frac{10}{15} = 0.67 \text{ [Hz]}$ (1) $v = \frac{2\pi r}{T} = \frac{2\pi(0.8)}{1.5} (1) [= 3.35 \text{ m s}^{-1}]$ Alternative: Total distance = $1.6\pi \times 10$ (1) $v = \frac{1.6\pi \times 10}{15} (1) [= 3.35 \text{ m s}^{-1}]$	1	1		2	2	
		(ii)		Substitution: $F = \frac{mv^2}{r} = \frac{(30 \times 10^{-3}) \times 3.35^2}{0.8} (1)$ $= 0.42 \text{ [N]}$ (1) Alternative: $F = m\omega^2 r = (30 \times 10^{-3}) \times 0.8 \times \left(\frac{2\pi}{1.5}\right)^2 (1)$ $= 0.42 \text{ [N]}$ (1)	1	1		2	2	
	(c)			Force = Tension + mg (1) Tension = Force – $mg = 0.42 \text{ ecf} - (30 \times 10^{-3})(9.81) = 0.13 \text{ [N]}$ (1) Award 1 mark for 0.7 [N]		2		2	1	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
(d)	(i)		Conservation of energy or implied (1) Full substitution or good algebra e.g. $\frac{1}{2}mv_b^2 = mgh + \frac{1}{2}mv^2$ OR Initial KE = 0.17 [J] and PE loss = 0.47 [J] (implied by final KE = 0.64 J) OR $v_b = \sqrt{2gh + v^2}$ (1) Final velocity = 6.53 [m s⁻¹] (1) Alternative: Use of $v^2 = u^2 + 2ax$ with $u = 3.35$ m s⁻¹ and $a = 9.81$ m s⁻² (1) Correct answer = 6.53 [m s⁻¹] (1) Some statement of why this equation works e.g. due to conservation of energy, we can consider a particle dropping with initial downward speed of 3.35 (1) Accept – this equation shouldn't work but it does! Alternative: Use of $v^2 = u^2 + 2ax$ with $u = 0$ gives $v = 5.6$ [m s⁻¹] (1) Pythagoras applied i.e. $\sqrt{3.35^2 + 5.6^2}$ (1) Correct answer = 6.53 [m s⁻¹] (1)		3		3	2	
	(ii)		Use of $x = ut + \frac{1}{2}at^2$ for the vertical motion or 2 other usable equations (1) $t = \sqrt{\frac{2x}{g}} = \sqrt{\frac{2(1.2+0.8)}{9.81}} = 0.64$ [s] (1) For the horizontal motion: horizontal distance from point of released = $vt = 3.35 \times 0.64$ ecf (1) = 2.14 [m] so claim is correct (1) ecf Accept 2.14 $\approx$ 2						
			Question 2 total	4	7	4	15	10	0

Vibrations

2017

6. (a) The pendulum in the figure below has a small bob of mass, m , suspended at the bottom of a light string of length, l . The string is shown at an angle, θ to the vertical and the mass, m , is at a distance, s , along the arc. The forces acting on the mass are shown.



- (i) Name the **two** forces acting on the mass. [1]

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- (ii) By considering these forces show that:

resultant force component on the mass along the arc = $-mg \sin \theta$

You may add to the diagram if you wish. [3]

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- (iii) If the oscillation is small so that $\sin \theta \approx \theta$ show in clear steps that the acceleration along the arc may be written as: [2]

$$\text{acceleration} = -\frac{gs}{l}$$

- (iv) Discuss whether the equation in part (a)(iii) satisfies the definition of simple harmonic motion. [2]

- (b) A small mass oscillates at the lower end of a pendulum of length 1.20 m.

- (i) Determine its:

I. period; [2]

II. frequency. [1]

- (ii) If the maximum displacement angle of the mass is 0.067 rad, justify the use of simple harmonic motion in part (b)(i). [2]

Question				Marking details				Marks available					
								AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		T tension in the string and mg weight of mass (gravitational force or gravity)				1			1		
		(ii)		T does not have a component tangential to the arc (1) Component of mg tangential to the arc is $mg \sin \theta$, (1) this is in the opposite direction to s (or θ) and so the negative sign (1)					3		3	1	
		(iii)		$\text{acceleration} = \frac{-mg \sin \theta}{m} = -g \sin \theta$ $= -g \theta \text{ (1) (using the approximation)}$ $\theta = \frac{s}{l} \text{ or } \sin \theta = \frac{s}{l} \text{ (1) } = -\frac{gs}{l}$					2		2	2	
		(iv)		Acceleration $\propto \theta$ (or s) measured [from a fixed position] (1) and opposite in direction (-ve) so SHM (1)						2	2		
	(b)	(i)	I	Substitution: $T = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{1.2}{9.81}}$ (1) $= 2.20 \text{ [s]}$ (1)				1	1		2	2	
			II	$f = \frac{1}{T} = 0.45 \text{ [5] [Hz]}$ ecf (1)					1		1	1	
		(ii)		For maximum distance along the arc $\theta_{\max} = 0.067$, also $\sin \theta_{\max} = 0.067$ As $\theta_{\max} = \sin \theta_{\max}$ (i.e. for the largest value of θ) (1) then $\sin \theta$ is equal to θ for all θ , and approximation holds for SHM. System oscillates with SHM (1)						2	2	1	
				Question 6 total				2	7	4	13	7	0

2018

6. (a) A load of 0.150 kg is attached to the lower end of a light vertical spring of stiffness (spring constant) 7.50 N m^{-1} . The system is initially in equilibrium. A student raises the load vertically through a distance of 0.095 m and releases it so that it oscillates.

Determine:

- (i) the extension of the spring when the system is in equilibrium; [2]

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- (ii) the period of oscillation. [2]

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- (b) The student notices the amplitude of oscillation decreasing and records the following amplitudes.

Oscillation number (n)	Amplitude (A) /m
0	0.095
10	0.062
20	0.043
30	0.029
40	0.019
50	0.014
60	0.009

She comments that "the rate of decrease of amplitude is larger at the start of the experiment than at its end". Justify the statement using the data from the table. [2]

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- (c) She expects that the measured amplitude of oscillation can be described by the equation:

$$A = A_0 e^{-\frac{n}{N}}$$

where A_0 is the initial amplitude, n is the oscillation number and N is a constant for the experiment that describes the decay of the amplitude.

- (i) Justify that A_0 is the amplitude when $n = 0$. [1]

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- (ii) By considering the amplitude when $n = 30$ use the equation to determine a value for N . [2]

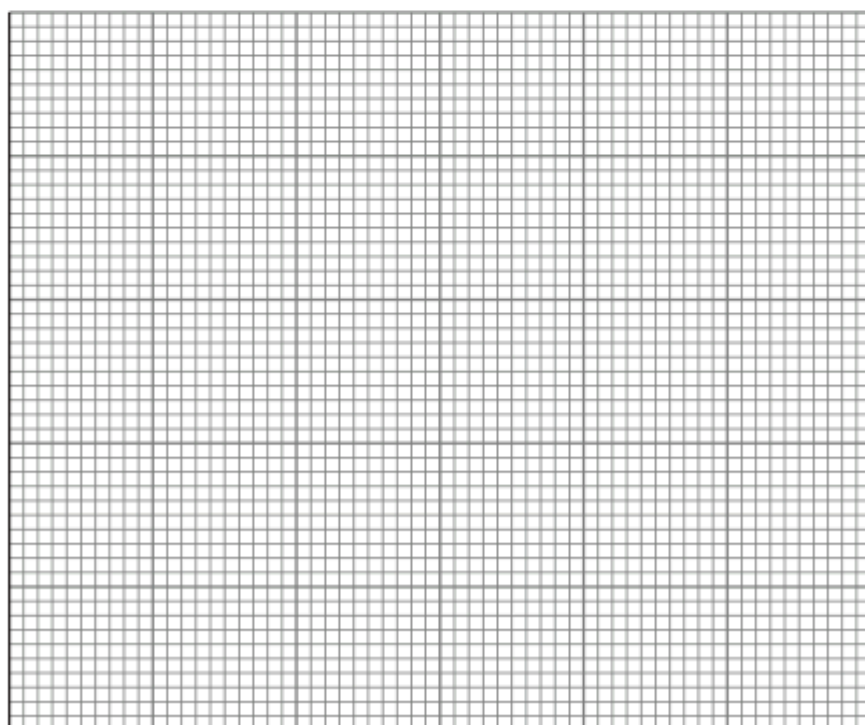
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(d) The student decides to check the validity of the equation.

- (i) Complete the third column in the table. The first four rows have already been completed. [1]

Oscillation number (n)	Amplitude (A)/m	$\ln\left(\frac{A_0}{A}\right)$
0	0.095	0
10	0.062	0.43
20	0.043	0.79
30	0.029	1.19
40	0.019	
50	0.014	
60	0.009	

- (ii) Plot $\ln\left(\frac{A_0}{A}\right)$ (y -axis) as a function of the oscillation number (x -axis) and draw a line of best fit. [3]



(iii) Explain whether or not your graph is in agreement with the equation $A = A_0 e^{-\frac{R}{N}}$. [2]

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(iv) Use the graph to determine a value for N . [3]

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(v) Explain which of the two values obtained for N is expected to be the more accurate, the value in part (c)(ii) or (d)(iv). [1]

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Question			Marking details	Marks available				Maths	Prac																								
				AO1	AO2	AO3	Total																										
6	(a)	(i)	$kx = mg$ (1) $x = \frac{0.150 \times 9.81}{7.5} = 0.196 \text{ m}$ answer (1)		2		2	2	2																								
		(ii)	$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.150}{7.5}}$ substitution (1) $= 0.889 \text{ s}$ (1) NB. No credit for use of $T = 2\pi\sqrt{\frac{l}{g}}$	1	1		2	2	2																								
	(b)		Drop in amplitude at beginning and end compared, e.g. 0.033 m and 0.005 m over first and last intervals. [Or rates of decrease: 0.0033 m per oscillation and 0.0005 m per oscillation] consider rate of decrease (1) Justify by noting the magnitude is larger at the start than at end of experiment. (1)			2	2		2																								
	(c)	(i)	When $n = 0$, $A = A_0 e^{\frac{0}{N}} = A_0 e^0 = A_0$ [Alternative: say $e^0 = 1$, so $A = A_0$.]			1	1	1	1																								
		(ii)	$A = A_0 e^{-\frac{n}{N}}$; $0.029 = 0.095 e^{-\frac{30}{N}}$ substitution (1) $N = \frac{-30}{\ln\left(\frac{0.029}{0.095}\right)}$ or equiv $\rightarrow N = 25.28 \approx 25$ answer (1)	1	1		2	2	2																								
	(d)	(i)	<table border="1"><thead><tr><th>Oscillation number (n)</th><th>Amplitude (A) /m</th><th></th></tr></thead><tbody><tr><td>0</td><td>0.095</td><td>0</td></tr><tr><td>10</td><td>0.062</td><td>0.43</td></tr><tr><td>20</td><td>0.043</td><td>0.79</td></tr><tr><td>30</td><td>0.029</td><td>1.19</td></tr><tr><td>40</td><td>0.019</td><td>1.61</td></tr><tr><td>50</td><td>0.014</td><td>1.91</td></tr><tr><td>60</td><td>0.009</td><td>2.36</td></tr></tbody></table> All three values correct (1) [accept 1.6, 1.9, 2.4 if consistent]	Oscillation number (n)	Amplitude (A) /m		0	0.095	0	10	0.062	0.43	20	0.043	0.79	30	0.029	1.19	40	0.019	1.61	50	0.014	1.91	60	0.009	2.36		1		1	1	1
Oscillation number (n)	Amplitude (A) /m																																
0	0.095	0																															
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40	0.019	1.61																															
50	0.014	1.91																															
60	0.009	2.36																															

		(ii)	Labelled axes with a suitable choice of scales (1) All points plotted correctly including origin (1) (ecf) Appropriate line of best fit [not necessarily through origin] ecf (1)	1	1		3	3	3
		(iii)	Any two of: 1) It is a straight line, or positive gradient (1) 2) Passes through the origin (1) 3) The measured points lie essentially on a straight line or little scatter (1)			2	2		2
		(iv)	Gradient = $\frac{2.35 - 0}{60 - 0}$ (from graph) = 0.039 ecf (1) $N = \frac{1}{\text{gradient}}$ (1) $= 25.6$ (1) Accept answers in range 25 – 26. Use of a point on the line – allow. Use of a data point not on the line $\rightarrow$ 1st mark not available.			3	3	3	3
		(v)	Expected to be the one from (d)(iv) as straight line based on all points [The value in (c)(ii) is calculated from only a single point, which may be anomalous.]			1	1		1
		Question 6 total		3	7	9	19	14	19

7. (a) A pendulum oscillating with simple harmonic motion consists of a small 0.05 kg mass oscillating at the end of a string of length 4.0 m. The displacement x in metres at time t can be written as:

$$x = 0.25 \cos \omega t$$

- (i) Show that the angular velocity, ω , is approximately 1.57 rad s^{-1} . [2]

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- (ii) Calculate the maximum speed of the mass. [2]

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- (iii) Show that the kinetic energy (E_k) of the mass, in joules, may be written as: [2]

$$E_k = 3.8 \times 10^{-3} \sin^2 1.57t$$

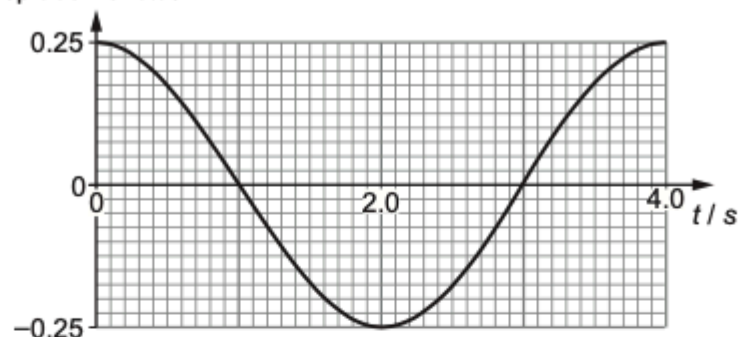
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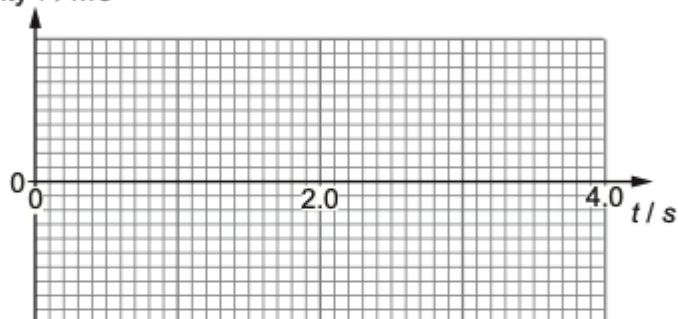
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- (iv) The displacement-time curve for the pendulum is shown for a time interval of one period i.e. 4.0 s. Draw curves for velocity, v , and E_k on the axes below for the same interval of time. Indicate values on your axes for v and E_k . [4]

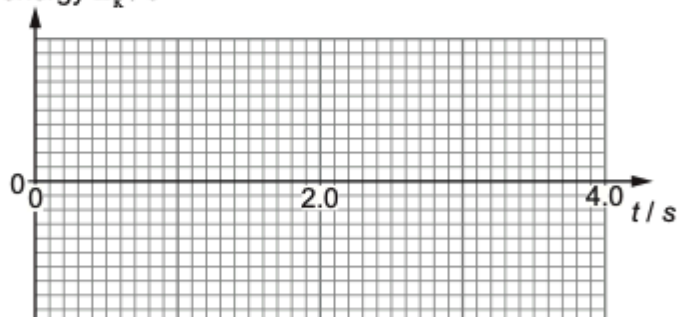
Displacement x / m



Velocity v / ms^{-1}



Kinetic energy E_k / J



- (v) Determine the maximum height reached by the pendulum mass above the level of its lowest point. [2]

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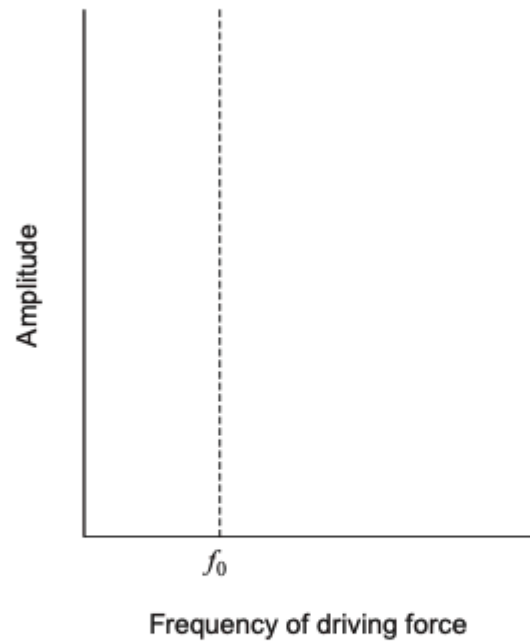
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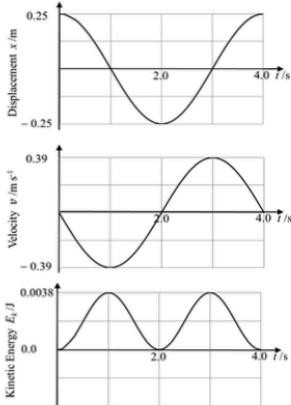
(b) A system that can oscillate may be driven by an external sinusoidal force.

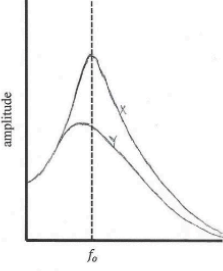
- (i) Sketch the amplitude of a lightly damped system as the frequency of the driver is increased. The natural frequency f_0 of the system is indicated. Label the curve **X**. [1]



- (ii) Name the effect seen near the frequency, f_0 . [1]

-
- (iii) Sketch a second curve on the **axes above** for an identical oscillating system that is damped to a greater extent. Label the curve as **Y**. [2]

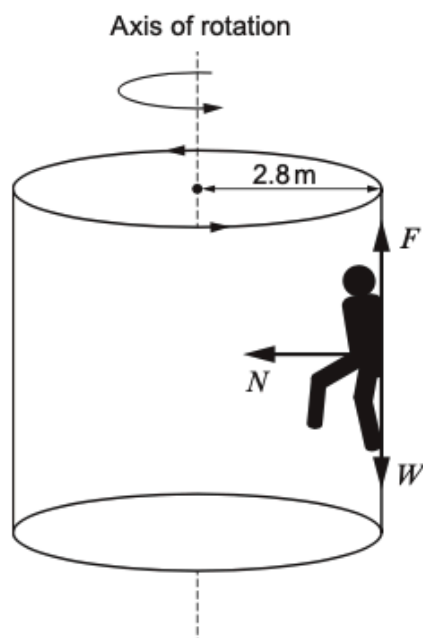
Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	$T = 2\pi\sqrt{\frac{l}{g}}$ $\omega = \frac{2\pi}{T} = \sqrt{\frac{g}{l}}$ combining formulae (1) = $\sqrt{\frac{9.81}{4.0}}$ $= 1.57 \text{ rad s}^{-1}$ (1) convincing		2		2	2	
		(ii)	$v_{\max} = \omega A$ (1) $v_{\max} [= 1.57 \times 0.25] = 0.39 \text{ m s}^{-1}$ (1)		2		2	1	
		(iii)	$E_k = \frac{1}{2}mv^2$ and $v = (-)A\omega \sin \omega t$ / $v = (-)0.39 \sin 1.57t$ (1) $E_k = \frac{1}{2} \times 0.05 \times (0.39)^2 \sin^2 1.57t$ (1) $= 3.8 \times 10^{-3} \sin^2 1.57t$ [convincing]	1	1		2	1	
		(iv)	 Phase of the velocity graph (1) Period of the kinetic energy graph (1) Kinetic energy curve always ≥ 0 with shape and phase (1)						

			Values on the axes of both graphs (1)		4		4	2	
		(v)	$mgh = 3.8 \times 10^{-3} \text{ J}$ ecf (1) $h = \frac{3.8 \times 10^{-3}}{0.05 \times 9.81} = 0.008 \text{ m}$ (1) (= 8 mm) Or: using trig: $4 - 4 \cos (0.25/4)$ (1) = 8 mm (1)		2		2	2	
(b)	(i)		 Sensible curve [X] with label [accept curve to the origin]	1			1		
	(ii)		Resonance	1			1		
	(iii)		Sensible curve with label (see part (b) (i)) [cannot cross X – always below, can co-incide at low frequencies] (1) Lower maximum amplitude at the same frequency or to left of maximum of curve X (1)	2			2		
			Question 7 total	5	11	0	16	8	0

2019

2. A fairground ride consists of a large empty cylinder of radius 2.80 m with its axis of rotation vertical. A person of mass 66.2 kg walks into the cylinder and stands against its inner wall. The cylinder is rotated around the axis. When the rotation rate is 36.0 revolutions per minute the floor is lowered, but the person remains fixed in place against the wall.

This is a student's diagram.



- (a) Define the *radian*.

[1]

- (b) Calculate:

- (i) the period of rotation, T ;

[2]

- (ii) the speed, v , at which the person is moving in the circle.

[2]

- (c) Three forces act on the person: the normal contact force from the wall, N , the weight, W , of the person, and the frictional force, F , from the wall. These are shown in the student's diagram.

Explain why the values of N and F are approximately 2600 N and 650 N respectively.

[4]

- (d) The maximum value that a frictional force can take is μN where μ is a dimensionless constant.

- (i) Show that the person does not slide down the wall provided that μ is larger than approximately 0.25. [2]

- (ii) A student says that as the cylinder slows down the floor will need to be raised to support the person when the angular velocity reduces below a certain value. Justify this and determine this angular velocity if μ has a value of 0.45. [4]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Angle when arc [length] equals radius Accept about 57.3° or angle when $2\pi = 360^\circ$ or cycle / circle	1			1		
	(b)	(i)	Use of $T = \frac{1}{f}$ (1) Answer = 1.67 [s] (1) Accept $\frac{5}{3}$ or 1.66 or 1.6̇ or 1.7 [s] Don't accept 1.6 [s]	1	1		2	1	
		(ii)	Substitution into $\omega = \frac{2\pi}{T}$ or $2\pi f$ or and $v = \omega r$ (1) ecf on T or f $v = 10.6 \text{ [m s}^{-1}\text{]} (1)$ (Accept 10.5 m s^{-1} if 1.67 s used)		2		2	2	
	(c)		$N = \frac{mv^2}{r}$ or $mr\omega^2$ or implied (1) $N = \frac{66.2 \times (10.6)^2}{2.8} = [2\ 634] \text{ [N]} (1)$ ecf on v and ω accept approximately 2 657 [N] $F = 66.2 \times 9.81 = [649.4 \text{ [N]}] (1)$ Vertical forces are balanced or equivalent e.g. $F = W$ (1)	1 1	1 1		4	2	
	(d)	(i)	$650 \leq \text{or} = \text{or} < \mu \times 2\ 600 (1)$ So $\mu > \text{or} = 0.24$ or $0.25 (1)$ Alternative: $2\ 600 \times 0.25 (1)$ $= 650 (1)$	1	1		2	2	
		(ii)	Friction = 650 [N] or implied (1) $\frac{650}{0.45} = 1\ 444 \text{ [N]} (1)$ Equating to centripetal (1) $\omega = 2.8 \text{ [rad s}^{-1}\text{]} (1)$ Answer of $2.51 \text{ [rad s}^{-1}\text{]}$ award 1 mark only		1	1 1 1	4	2	
			Question 2 total	5	7	3	15	9	0

3. Simple harmonic motion (SHM) may be represented by:

$$a = -\omega^2 x$$

(a) State what quantities are represented by: [1]

a

ω

x

(b) If the amplitude of the oscillation of an object moving with SHM is 0.012m and the period is 0.40 s:

(i) determine the maximum acceleration; [3]

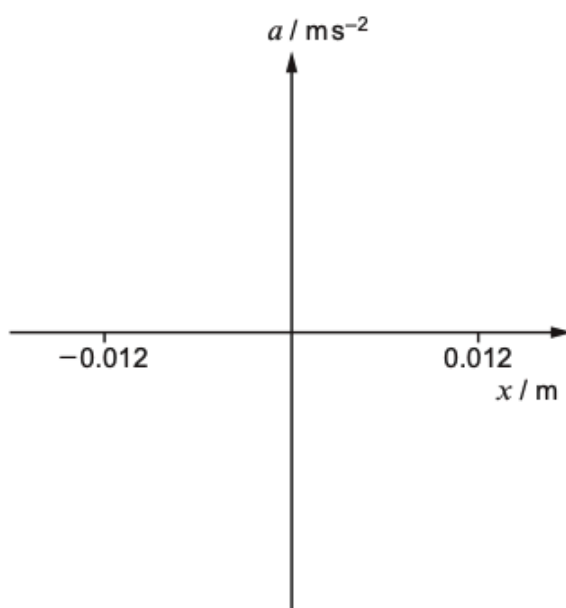
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(ii) sketch a graph showing the variation of a with x . Insert values on the a -axis. Values are given on the x -axis. [3]



- (c) The oscillating object in part (b) is at its centre position when $t = 0$ and moving in the positive x direction. Insert the missing values in the expression for x in metres. [3]

$$x = \boxed{} \cos(\boxed{} t + \boxed{})$$

- (d) An oscillating system may be driven by an external force. Describe and explain:

- (i) an application of forced oscillations that is useful; [2]

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- (ii) an example of forced oscillations that should be avoided. [3]

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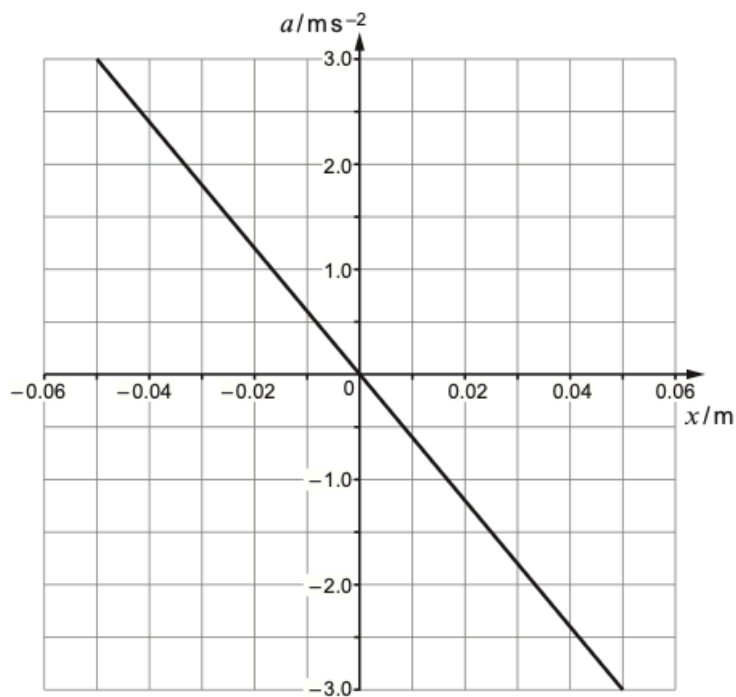
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		a = acceleration ω = angular velocity or angular frequency or pulsance x = displacement All 3 correct (1)	1			1		
	(b)	(i)	$\omega = \frac{2\pi}{0.4} (1) [= 15.7 \text{ s}^{-1}]$ $a_{\max} = \omega^2 A = (15.7)^2 \times 0.012 (1)$ $= 2.96 [\text{m s}^{-2}] (1)$	1	1 1		3	2	
		(ii)	One value of a on graph (1) ecf from (b)(i) Any straight line (1) Correct position of line (1)		3		3	2	
	(c)		$x = \boxed{0.012} \cos\left(\boxed{15.7} t + \boxed{\frac{3\pi}{2}}\right)$ (1 $\times$ 3 – one mark for each box) (alternative for angle: $-\frac{\pi}{2}$) Accept 5π for 15.7. ecf on ω		3		3	1	
	(d)	(i)	Example e.g. microwave ovens or swing (1) Oscillator and driving force named e.g. water molecules and microwaves or swing and person pushing (1)	2			2		
		(ii)	Example and consequence e.g. bridge and falling or something in the dashboard and buzzing (1) Driving force e.g. wind / soldiers marching or engine (1) Resonance explained i.e. both frequencies are the same (1)	3			3		
			Question 3 total	7	8	0	15	5	0

4. Describe an experiment that uses a pendulum to determine the acceleration due to gravity with graphical analysis. [6 QER]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4			Measurement: Diagram to assist answer Measure length of string Measure time for several oscillations Use of fiducial point Repeat for each length Repeat for several lengths of string Analysis and theory: Small angle / amplitude Use of $T = 2\pi\sqrt{\frac{l}{g}}$ Plot T^2 vs l or T vs $\sqrt{l}$ Is a straight line [through the origin] Gradient is $\frac{4\pi^2}{g}$ or $\frac{2\pi}{\sqrt{g}}$ respectively Use the gradient or points to calculate g	6			6		6

2022

3. A small object of mass 0.30 kg oscillates at the end of a spring. The graph shows how its acceleration, a , depends on its displacement, x , from a fixed point.



- (a) (i) Identify the **two** characteristics of the graph that show that the object is oscillating with *simple harmonic motion*. [2]

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- (ii) Use the graph to show that the angular frequency, ω , of the oscillation is approximately 7.7 rad s^{-1} . [3]

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(iii) Calculate:

I. the period of oscillation;

[1]

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II. the spring constant;

[2]

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III. the velocity of the object as it passes through the centre of oscillation.

[2]

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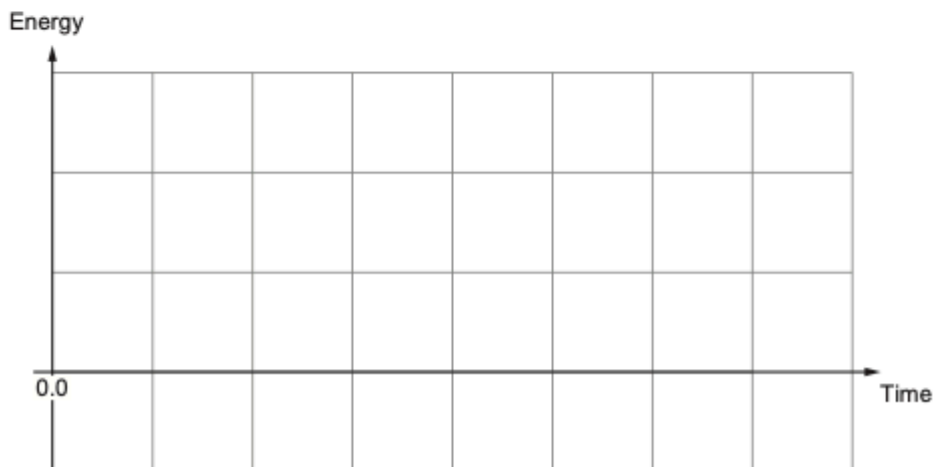
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(iv) The object was released at the displacement of -0.05 m at time $t = 0$. Sketch on the axes below:

- I. the potential energy of the system (label the sketch PE),
- II. the kinetic energy of the system (label the sketch KE).

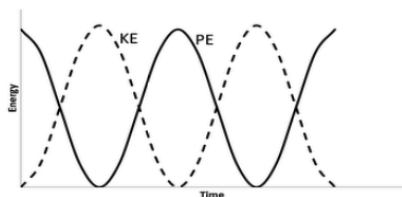
Draw both sketches **over one cycle** of the undamped oscillation. (No values are required on the axes.)

[3]



- [6 QER]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Straight line <u>through the origin</u> [shows that acceleration is directly proportional to displacement] (1) Negative slope [shows that the acceleration is always directed towards the fixed point] (1)	2			2		
		(ii)	Calculating gradient i.e. 60 or obtaining suitable point from graph e.g. -0.05, 3.0 (1) Gradient = [-] ω^2 or substitution into equation (1) $\omega = 7.75 \text{ [rad s}^{-1}\text{]} (1)$ Allow 1 mark for slips such as: $\omega = \sqrt{\frac{3}{0.5}}$ Allow 2 marks for $\omega = \sqrt{\frac{3}{0.5}} = 2.45$ or $\omega = \sqrt{\frac{1.5}{0.03}} = 7.07$ Allow 3 marks for Accept $\sqrt{\frac{1.75}{0.03}} = 7.64 \text{ [rad s}^{-1}\text{]} \text{ or } \sqrt{\frac{2}{0.033}} = 7.8 \text{ [rad s}^{-1}\text{]} \text{ or } \sqrt{\frac{2.4}{0.04}} = 7.85 \text{ [rad s}^{-1}\text{]}$ Allow 1 mark for: $\omega = \sqrt{\frac{a_{\max}}{A}} = 7.7$ i.e. no evidence of data points used or substitution but method ok. Alternative for full marks: Using $a = \frac{v^2}{r}$ (which leads to $v = 0.39$) (1) Using $\omega = \frac{v}{r}$ (1) Convincing correct answer e.g. 7.75 or $\frac{0.39}{0.05}$ (1)		3		3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii) I.	$T = \frac{2\pi}{\omega} = \frac{2\pi}{7.7} = 0.82 \text{ [s]}$		1		1	1	
		II.	Substitution: $T = 2\pi\sqrt{\frac{m}{k}}$ so $0.82 = 2\pi\sqrt{\frac{0.30}{k}}$ or $ma = kx$ (1) $k = 17.8 \text{ N m}^{-1}$ unit mark (1)	1	1		2	2	
		III.	Substitution: $v = \omega A = 7.7 \times 0.05$ (1) if $\sin\omega t$ or $\cos\omega t$ used then the angle must be correct $v = 0.39 \text{ [m s}^{-1}\text{]} (1)$ Alternatives for the first mark: Substituting into $ma = kx$ e.g. $0.3 \times 3 = k \times 0.05$ (1) OR substituting into $\omega = \sqrt{\frac{k}{m}}$ e.g. $7.8 = \sqrt{\frac{k}{0.3}}$ (1)	1	1		2	2	
		(iv)	 Shape and start point of PE sketch (1) Shape KE sketch and starting at zero (1) KE out of phase with PE and same size peaks and one complete cycle i.e. 2 KE peaks (1)		3		3	1	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		Indicative content: <u>Damping:</u> Damping linked to resistive force. It dissipates energy or linked to energy loss. Amplitude decreases with time. Link to energy loss to surroundings. Critical damping. <u>Resonance:</u> Periodic force. Happens when frequency is equal to the natural frequency. Amplitude of oscillation becomes large. Damping lowers the resonance peak. Reasonable resonance graph. <u>Real systems:</u> Any sensible real system e.g. bridges / structural / swings, tuning circuit Example of damping e.g. car suspension With a description of the driving force Explanation of whether resonance is good or bad.	6			6		

2023

6. A student says that a simple pendulum oscillates with simple harmonic motion with a period, T , of:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum and g is the acceleration due to gravity.

- (a) Describe what is meant by a simple pendulum.

[1]

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- (b) State what is meant by simple harmonic motion.

[2]

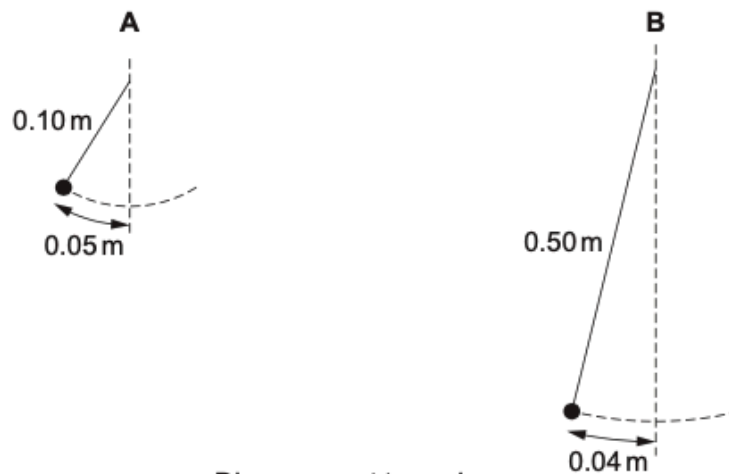
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- (c) The student sets up two experimental systems to investigate simple harmonic motion. System A is a pendulum of length 0.10 m oscillating with an amplitude of 0.05 m. System B is a pendulum of length 0.50 m oscillating with an amplitude of 0.04 m. Give two reasons why system B is better than system A to investigate simple harmonic motion.

[2]



Diagrams not to scale.

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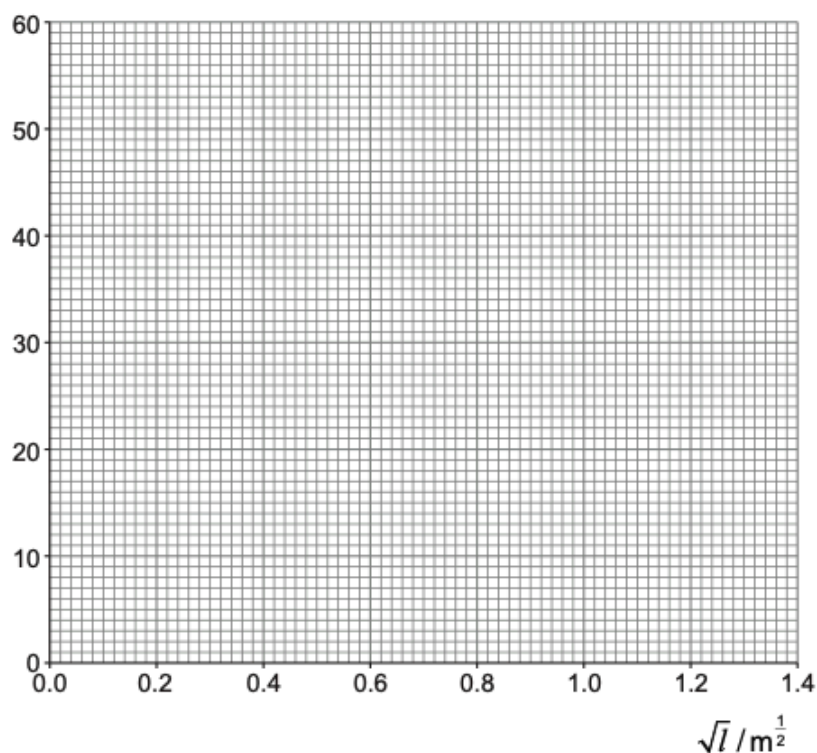
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- (d) Measurements for an oscillating simple pendulum are given in the table below, where l is the length of the pendulum and T_{20} is the time taken for 20 oscillations. For each value of l four values of T_{20} are measured.

l/m	$\sqrt{l}/\text{m}^{\frac{1}{2}}$	T_{20}/s	Mean T_{20}/s	Uncertainty T_{20}/s
0.250	0.50	19, 21, 22, 20	21	2
0.500	0.71	26, 29, 26, 29	28	2
0.750		35, 34, 33, 36		
1.000		42, 39, 40, 41		
1.250		45, 42, 44, 45		
1.500		48, 50, 49, 52		

- (i) **Complete the table** by determining $\sqrt{l}$, the mean time for 20 oscillations, T_{20} and the uncertainty in T_{20} . Values have been inserted for the first two lengths. [3]
- (ii) Plot mean T_{20} against $\sqrt{l}$ on the grid below. Include error bars on the T_{20} axis only. Draw a line of maximum gradient and a line of minimum gradient. [4]

Mean T_{20}/s



(iii) Justify why error bars for $\sqrt{t}$ are not plotted on the graph. [2]

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(iv) Use the maximum and minimum gradients to determine a mean value for the acceleration due to gravity, g , and its **percentage** uncertainty. [5]

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(e) Another student says that the gradient of the graph would be approximately twice as large if the experiment was carried out on the Moon, where the acceleration is approximately $0.2g$. Investigate if this statement is correct, justifying your answer. [3]

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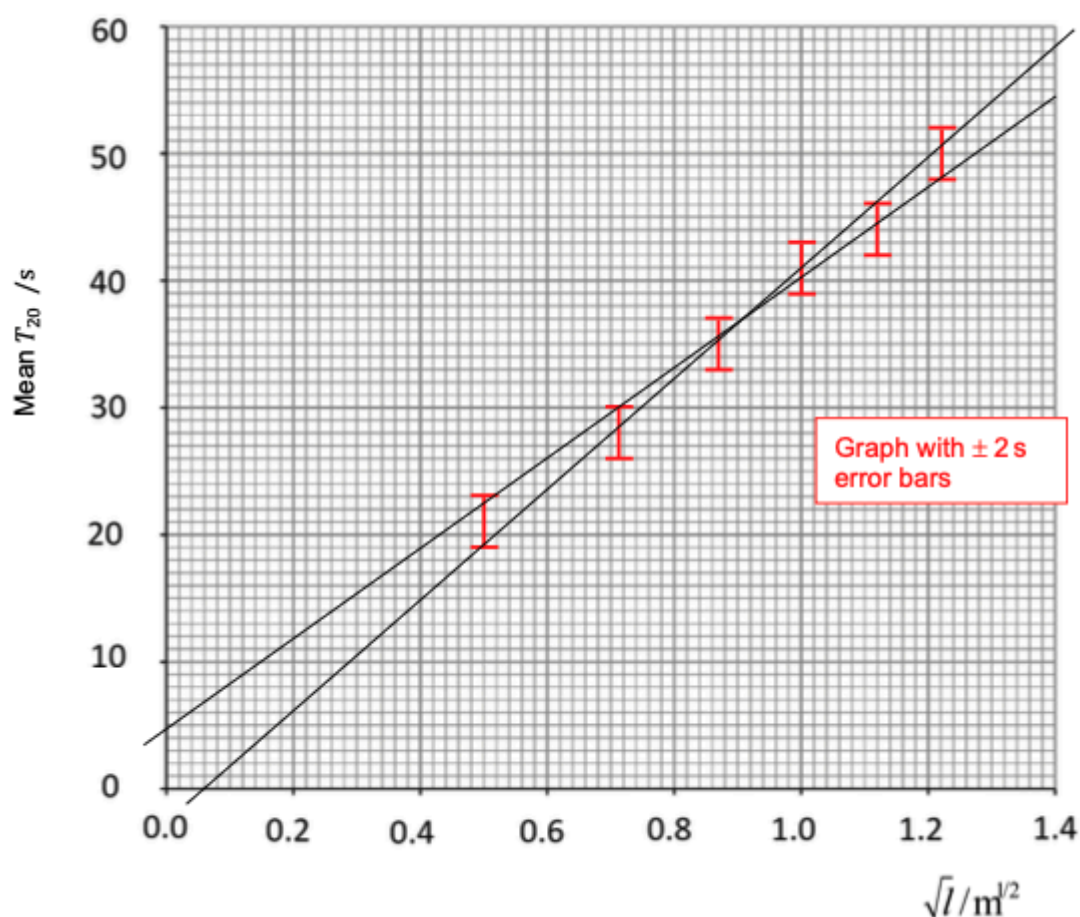
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Question				Marking details	Marks available																																								
					AO1	AO2	AO3	Total	Maths	Prac																																			
6	(a)			Is a [light] {string / thread} with a {bob / mass / weight} attached to one end	1			1		1																																			
	(b)			Acceleration is proportional to its {distance / displacement} (1) and directed towards its {equilibrium / fixed} point (1)	2			2																																					
	(c)			[System A] the angle is too big [for shm to occur] (1) Any × (1) from: - the shorter length {has a larger [percentage] uncertainty / is less accurate} - the shorter period {has a larger [percentage] uncertainty / is less accurate}			2	2		2																																			
	(d)	(i)		<table><tr><th>l / m</th><th>$\sqrt{l}$ / m^{1/2}</th><th>T_{20} / s</th><th>Mean T_{20} / s</th><th>Uncertainty T_{20} / s</th></tr><tr><td>0.250</td><td>0.50</td><td>19, 21, 22, 20</td><td>21</td><td>2</td></tr><tr><td>0.500</td><td>0.71</td><td>26, 29, 26, 29</td><td>28</td><td>2</td></tr><tr><td>0.750</td><td>0.87</td><td>35, 34, 33, 36</td><td>35</td><td>2</td></tr><tr><td>1.000</td><td>1.00</td><td>42, 39, 40, 41</td><td>41</td><td>2</td></tr><tr><td>1.250</td><td>1.12</td><td>45, 42, 44, 45</td><td>44</td><td>2</td></tr><tr><td>1.500</td><td>1.22</td><td>48, 50, 49, 52</td><td>50</td><td>2</td></tr></table> [1 for each column] Deduct a maximum of 1 mark overall for inconsistent sig figs in each column	l / m	$\sqrt{l}$ / m ^{1/2}	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s	0.250	0.50	19, 21, 22, 20	21	2	0.500	0.71	26, 29, 26, 29	28	2	0.750	0.87	35, 34, 33, 36	35	2	1.000	1.00	42, 39, 40, 41	41	2	1.250	1.12	45, 42, 44, 45	44	2	1.500	1.22	48, 50, 49, 52	50	2		3		3	3	3
l / m	$\sqrt{l}$ / m ^{1/2}	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s																																									
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1.250	1.12	45, 42, 44, 45	44	2																																									
1.500	1.22	48, 50, 49, 52	50	2																																									



Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		See example graphs at end of Q6 Award 3 marks for all points and error bars correct Award 2 marks for 5 points and error bars correct Award 1 mark for 4 points and error bars correct Award 0 marks for 3 points or less and error bars correct Correct lines drawn for the maximum gradient and the minimum gradient (1) ecf		4		4	4	4
		(iii)		l can be measured to ± 0.001 [m] or resolution is 1 mm (1) [Uncertainty in $\sqrt{l}$] is too small to plot (1)			2	2		2
		(iv)		Both gradients correct (1) Mean gradient correct (1) ecf Absolute uncertainty in gradient correct (1) ecf implied by % uncertainty Correct g (1) ecf Final % correct (1) ecf No sig fig or unit penalties $m_{\max} \left[= \frac{58.2}{1.40 - 0.06} \right] = 43.4; m_{\min} \left[= \frac{54.3 - 4.7}{1.4} \right] = 35.4$ $\therefore \text{Gradient, } m = 39.4 \pm \{4.0 / 10\% \}$ Either For 20 oscillations, $T = 20 \times 2\pi \sqrt{\frac{l}{g}}$ $\therefore \sqrt{g} = \frac{40\pi}{m} \text{ [or by impl.] } [= 3.19 \pm 10\%]$ $\therefore g = 10.2 \text{ m s}^{-2} \pm 20\%$		5		5	5	5

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				Or For one oscillation: $m = 1.97 \pm 10\%$ (1) $\left[T = 2\pi \sqrt{\frac{l}{g}} \right], \therefore \sqrt{g} = \frac{2\pi}{m} \text{ [or by impl.] } [= 3.19 \pm 10\%] \text{ (1)}$ $\therefore g = 10.2 (\pm 20\%) \text{ m s}^{-2}$ Alternative: calculate max value of g from min gradient; min value of g from max gradient and hence $g \pm$ uncertainty. Gradients of both: 43.284 and 35.357 (1) Mean $T_{20} = \frac{40\pi}{\sqrt{g}} \sqrt{l}$ so gradient = $\frac{40\pi}{\sqrt{g}}$ $g = \left(\frac{40\pi}{43.3} \right)^2 = 8.42 \text{ m s}^{-2} \text{ and}$ $g = \left(\frac{40\pi}{35.3} \right)^2 = 12.67 \text{ m s}^{-2} \text{ (1)}$ So $\text{unc}(g) = \frac{12.67 - 8.42}{2} = 2.1 \text{ m s}^{-2} \text{ (1)}$ $g = 10.5 \text{ m s}^{-2} [1]$ $\therefore \% \text{ unc}(g) = \left[= \frac{2.1}{10.5} \times 100\% \right] = 20\% \text{ (1)}$						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			Slope $\propto \frac{1}{\sqrt{g}}$ or equivalent e.g. gradient = $\frac{2\pi}{\sqrt{g}}$ or gradient = $\frac{40\pi}{\sqrt{g}}$ (1) Ratio of slopes = $\frac{1}{\sqrt{0.2}}$ or $\sqrt{5}$ or new gradient = 89 or two correct expressions for gradients e.g. $\frac{2\pi}{\sqrt{g}}$ and $\frac{2\pi}{\sqrt{0.2g}}$ or $\frac{40\pi}{\sqrt{g}}$ and $\frac{40\pi}{\sqrt{0.2g}}$ (1) Ratio of gradients = 2.24ish and sensible comment (1)			3	3	3	
				Question 6 total	3	12	7	22	15	17

Kinetic Theory

2017

4. (a) Heat is supplied to an ideal gas in a sealed container of fixed volume. Explain carefully the physical processes that cause an increase in the **temperature and pressure** of the gas. [6 QER]

(b) A container of fixed volume contains oxygen gas at a temperature of 293 K.

- (i) Five oxygen molecules have speeds 400, 425, 450, 550 and 625 m s⁻¹. Determine their rms speed. [2]

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- (ii) Using an appropriate calculation, determine whether or not the rms speed calculated in part (b)(i) is consistent with the expected rms speed of the molecules of the gas at this temperature. (Relative molecular mass of oxygen gas = 32.) [3]

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- (iii) If the gas in the container is heated and the pressure of the gas increases by 20% of its initial value, determine the rms speed of the molecules of the gas. [2]

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Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
4	(a)	Increase in temperature - kinetic theory effects Molecules move randomly Collisions become more frequent [when heat supplied] [No change in volume so heat flowing in] causes increase in U / kinetic energy No work done [as constant volume] T increases with U as temperature proportional to U, or equivalent Newton's laws of motion Momentum of molecules increase Force on molecules = rate of change of momentum (during collision with wall) Force on wall is equal and opposite to force on molecules Greater forces during the collisions Increase in pressure Molecules collide with walls exerting force on walls and / or pressure Pressure increases with temperature Pressure = force on walls per unit area Mean pressure due to many collisions [and many molecules] 5-6 marks Comprehensive account including reference to increase in temperature, Newton's laws of motion and increase in pressure. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Comprehensive account including reference to 2 out of 3 of increase in temperature, Newton's laws of motion and increase in pressure or brief account of all 3 areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			1-2 marks Comprehensive account including reference to one of increase in temperature, Newton's laws of motion and increase in pressure or limited account of 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
(b)	(i)	$\text{rms} = \sqrt{\frac{400^2+425^2+450^2+550^2+625^2}{5}} \quad (1)$ $= 497 \text{ [ms}^{-1}\text{]} \quad (1)$		2		2	2		
	(ii)	The expected rms is explained by: $PV = nRT \quad \text{so} \quad P = \frac{nRT}{V}$ $\rho = \text{mass}/V = \frac{n(M_r \times 10^{-3})}{V} \quad \text{use of both equations by substitution (1)}$ substitute these into: $P = \frac{1}{3}\rho\overline{c^2}$; $\sqrt{\overline{c^2}} = \sqrt{\frac{3P}{\rho}}$ $\sqrt{\overline{c^2}} = \sqrt{3 \frac{nRT}{V} \frac{V}{n(M_r \times 10^{-3})}} = \sqrt{\frac{3RT}{(M_r \times 10^{-3})}}$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3(8.31)(293)}{(32 \times 10^{-3})}} = 478 \text{ [ms}^{-1}\text{]} \quad (1)$ Valid conclusion with data i.e. yes, the rms speed of the five molecules is slightly higher [about 4% above the expected rms of the gas] (1) ecf Alternative solution: Use $m\overline{c^2} = 3kT$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(293)}{(32)(1.66 \times 10^{-27})}}$ with sensible substitution (1)= 478 [ms⁻¹] (1)		1	1	3	3		

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
		(iii)	Density does not change and Pressure increases from the original pressure p to $1.2p$ (1) i.e. an increase of 20%. so $(\sqrt{c^2})_{\text{new}} = \sqrt{\frac{3(1.2p)}{\rho}}$ $(\sqrt{c^2})_{\text{new}} = \sqrt{1.2} \sqrt{\frac{3p}{\rho}} = \sqrt{1.2} (478) = 524 \text{ m s}^{-1} \quad (1)$		2		2	2	
			Question 4 total	6	5	2	13	7	0

2018

2. Kinetic theory for an ideal gas gives an expression that can be written as:

$$pV = \frac{1}{3} nN_A m \overline{c^2}$$

- (a) State the meaning of the terms: [2]

(i) m

(ii) $\overline{c^2}$

- (b) Defining both symbols, explain what quantity is given by nN_A . [3]

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- (c) The product of pressure and volume for an ideal gas may also be expressed as $pV = nRT$. Show in clear steps that the total translational kinetic energy of one mole of the gas is $\frac{3}{2}RT$. [4]

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- (d) An ideal monatomic gas, initially at a pressure of 115 kPa and temperature of 294 K, expands at constant pressure from a volume of $2.20 \times 10^{-3} \text{ m}^3$ to $2.60 \times 10^{-3} \text{ m}^3$. Calculate the change in the internal energy of the gas. [3]

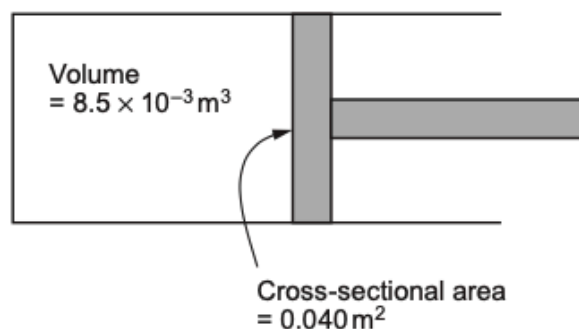
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i) (ii)	Mass of a molecule [accept: atom or particle] [of the gas] Mean [accept: average] square velocity / speed [of a molecule of the gas]	1 1			2		
	(b)		N_A [Avogadro's number is] the number of molecules per mole [or in a mole] [of gas] (1) There are n moles [of the gas] (1) So the total number of molecules [in the gas] = $n N_A$ (1)	3			3		
	(c)		Realising that $\text{KE} = N_{(A)} \frac{1}{2} \overline{mc^2}$ (1) Substitution of $pV = nRT$ (1) Convincing algebra, e.g. handling $\frac{2}{3} \frac{1}{3} \frac{3}{2}$ (1) Explanation that $n = 1$ [NB. if M used need to see molar mass when $n = 1$] (1) Alternative: $\frac{1}{2} \overline{mc^2} = \frac{3}{2} kT$ (1) Multiplication by $N \rightarrow \frac{1}{2} N \overline{mc^2} = \frac{3}{2} NkT$ (1) Justification that $R = kN_A$ (1) with justification (1)	4			4	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)		$n = \frac{pV}{RT} = \frac{115 \times 10^3 \times 2.2 \times 10^{-3}}{8.31 \times 294} = 0.104 \text{ [mol]} (1)$ Final temperature, $T = \frac{pV}{nR} = \frac{115 \times 10^3 \times 2.6 \times 10^{-3}}{0.104 \times 8.31} = 346.0 \text{ K} (1)$ [value depends upon rounding of n . No rounding $\rightarrow 347.5 \text{ K}$] $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} \times 0.104 \times 8.31 (346 - 294) = 67.4 \text{ J} (1)$ Allow ecf on 1 mol for the last two marks. Alternative: $\Delta U = \frac{3}{2} (p_1 V_1 - p_2 V_2)$ or $\Delta U = \frac{3}{2} p \Delta V$ (for constant pressure) (1) $= \frac{3}{2} 115 \times 10^3 (2.60 - 2.20) \times 10^{-3} (1)$ $= 69 \text{ J} (1)$		3		3	3	
			Question 2 total	9	3	0	12	5	0

2019

5. A sealed container with a leakproof piston of cross-sectional area 0.040 m^2 at one end has an initial volume of $8.5 \times 10^{-3}\text{ m}^3$. It contains oxygen gas at a pressure of $5.0 \times 10^5\text{ Pa}$ and a temperature of 285 K .

(Relative molecular mass of oxygen = 32)



(a) Calculate:

- (i) the number of moles of oxygen gas in the container;

[2]

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- (ii) the number of molecules in the container;

[1]

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- (iii) the rms speed of the molecules;

[3]

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(iv) the force exerted by the gas on the piston.

[1]

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(b) The gas is allowed to expand at a constant temperature of 285 K to a volume of $10.2 \times 10^{-3} \text{ m}^3$.

(i) Calculate the final pressure of the gas.

[2]

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(ii) During the process the gas does 773 J of work. Determine the heat flowing into the gas, explaining your reasoning. [2]

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(iii) Following the expansion the gas is returned to its initial state by a two-stage process:

- a decrease in volume at constant pressure;
- an increase in pressure at constant volume.

A student says that approximately 60 J of heat flows into the gas during the overall cycle (the expansion and return of the gas to its original state). Justify this. Include a sketch of the paths on a p – V diagram with values on both axes. [4]

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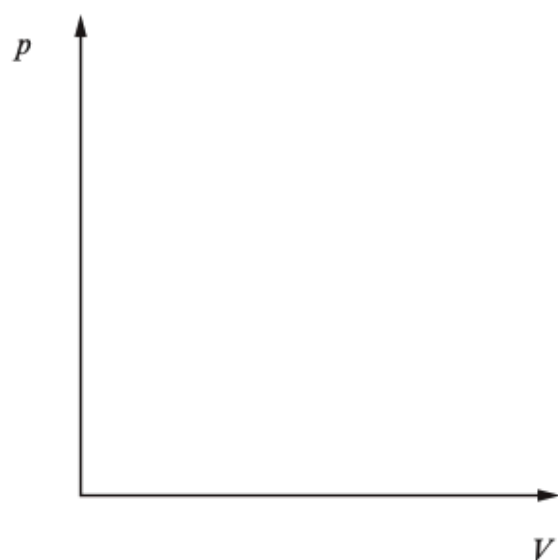
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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	Substitution $n = \frac{pV}{RT} = \frac{(5 \times 10^5)(8.5 \times 10^{-3})}{(8.31)(285)}$ (1) $= 1.79 \text{ [mol]} \text{ (1)}$	1	1		2	2	
		(ii)	$N = N_A n = (6.02 \times 10^{23}) \times 1.79 \text{ ecf} = 1.08 \times 10^{24}$		1		1	1	
		(iii)	Substitution of p and V or k and T (1) Correct use of Nm or m in either: $p = \frac{1}{3} \rho c^2$ or $pV = \frac{1}{3} Nmc^2$ ecf (1) $c_{rms} = 471 \text{ [ms}^{-1}]$ (1)	1	1 1		3	2	
		(iv)	Force $= pA = (5.0 \times 10^5) \times 0.04 = 20\,000 \text{ [N]}$		1		1	1	
	(b)	(i)	Substitution e.g. $p = \frac{5.0 \times 10^5 \times 8.5 \times 10^{-3}}{10.2 \times 10^{-3}}$ (1) ecf on n if $pV = nRT$ used $p = 420 \text{ k[Pa]} \text{ (1)}$	1	1		2	2	
		(ii)	$\Delta U = 0$ (1) So by using the first law of thermodynamics $\Delta U = Q - W$ hence $Q = W = 773 \text{ [J]} \text{ (1)}$	1					
					1		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Work done $= [-]710 \text{ [J]}$ or area of triangle attempted (1) Total work done by the gas around cycle $= 773 - 710 + 0 = 63 \text{ [J]}$ and $Q = W = 63 \text{ [J]} \text{ (1)}$ Axes labelled with units (1) Correct closed triangle as shown (1) Treat arrows as neutral $p / 10^5 \text{ Pa}$ $V / 10^{-3} \text{ m}^3$			4	4	3	
			Question 5 total	4	7	4	15	13	0

2022

2. A balloon of volume $1.60 \times 10^{-3} \text{ m}^3$ contains helium at a pressure of $1.20 \times 10^5 \text{ Pa}$ and a temperature of 293 K . (Relative molecular mass of helium = 4.)

(a) Determine:

- (i) the number of molecules in the balloon; [2]

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- (ii) the rms speed of the molecules. [2]

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- (b) The product of the pressure and the volume of the gas is given by $pV = nRT$ and also by $pV = \frac{1}{3}Nmc^2$.

- (i) Show that the mean kinetic energy of a gas molecule is $\frac{3}{2}kT$ where k is the Boltzmann constant. [4]

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- (ii) Calculate the mean kinetic energy of a molecule in the gas. [1]

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Thermal

- (iii) Determine the internal energy of the gas in the balloon. [1]

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- (c) (i) Explain, in terms of the molecules, the difference in the nature of the **internal energy** in an ideal gas and in a liquid. [3]

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- (ii) State the temperature of a system for its internal energy to be a minimum. [1]

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	$n = \frac{pV}{RT} = \frac{1.2 \times 10^5 \times 1.6 \times 10^{-3}}{8.31 \times 293.0} = [0.0789 \text{ mol}] \text{ (1)}$ $nN_A = 0.0789 \times 6.02 \times 10^{23} = 4.75 \times 10^{22} \text{ (1)}$ Alternative: $N = \frac{pV}{kT} = \frac{1.2 \times 10^5 \times 1.6 \times 10^{-3}}{1.38 \times 10^{-23} \times 293.0} \text{ (1)}$ $= 4.75 \times 10^{22} \text{ (1)}$		2		2	2	
		(ii)	Valid method e.g. $\frac{1}{2}mv^2 = \frac{3}{2}kT \text{ (1)}$ Answer = 1 350 [m s ⁻¹] ecf on $N \text{ (1)}$	1	1		2	2	
	(b)	(i)	Combine: $nRT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ Substitution of $N = nN_A$ or $n = 1$ so $N = N_A \text{ (1)}$ KE [of a molecule] = $\frac{1}{2}m\overline{c^2}$ or $U = N\frac{1}{2}m\overline{c^2}$ can be implied if everything is clear (1) So use or implied $k = \frac{R}{N_A}$ and convincing algebra (1) Common alternative for full marks: $nRT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ $NkT = \frac{1}{3}Nm\overline{c^2} \text{ (1) (implies 1st line if 1st line omitted)}$ $KE = \frac{1}{2}m\overline{c^2} \text{ (1) (can be implied if everything clear)}$ Algebra leading to $KE = \frac{3}{2}kT \text{ (1)}$	4			4	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Another alternative for full marks: $KE = \frac{1}{2}m\overline{c^2} \text{ (1)}$ Substituting from $pV = \frac{1}{3}Nm\overline{c^2}$ into above e.g. $KE = \frac{1}{2} \frac{3pV}{N} \text{ (1)}$ Substituting $pV = NkT$ into above e.g. $KE = \frac{1}{2} \frac{3NkT}{N} \text{ (1)}$ Hence $KE = \frac{3}{2}kT \text{ (1)}$ Alternative for a maximum of 3 marks: Internal energy or $U = \frac{3}{2}nRT \text{ (1)}$ $U = \frac{3}{2}NkT \text{ (1)}$ This is for N molecules, hence $KE = \frac{3}{2}kT \text{ (1)}$						
		(ii)	$\frac{3}{2} \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21} \text{ [J]}$		1		1	1	
		(iii)	$4.75 \times 10^{22} \text{ ecf} \times 6.07 \times 10^{-21} \text{ ecf} = 288 \text{ [J]}$		1		1	1	
	(c)	(i)	Ideal gas – kinetic energy only or no PE (1) Liquid – kinetic energy and potential energy (1) Link PE to intermolecular forces or separation of molecules or intermolecular force / separation mentioned for liquid (1)	3			3		
		(ii)	At absolute zero or 0 [K] or -273 [°C] Do not accept 0 °C OR -273 K (treat as neutral, don't penalise – they may have said something else that is correct e.g. absolute zero)	1			1		
			Question 2 total	9	5	0	14	8	0

Thermal

2017

3. (a) The first law of thermodynamics is given by:

$$\Delta U = Q - W$$

State what is represented by:

[3]

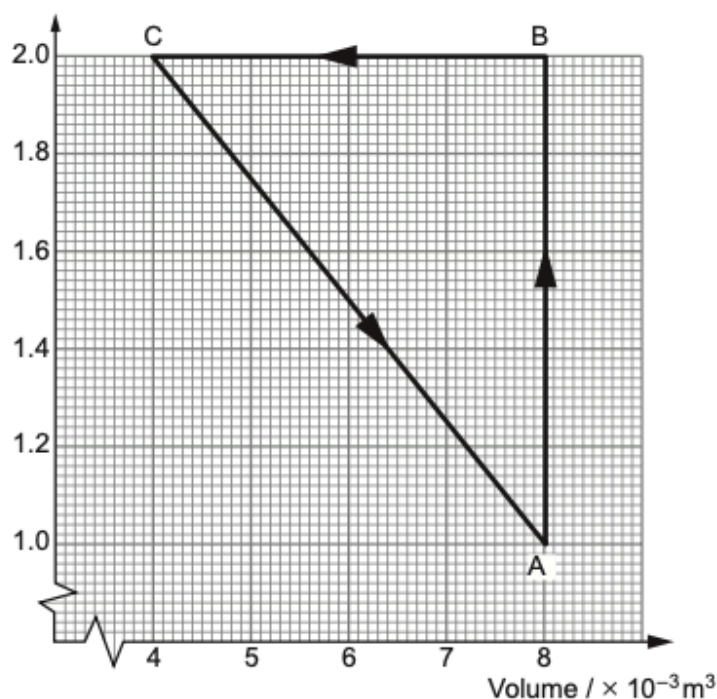
ΔU

Q

W

- (b) A fixed mass of gas is taken around the closed cycle $A \rightarrow B \rightarrow C \rightarrow A$.

Pressure / $\times 10^5 \text{ N m}^{-2}$



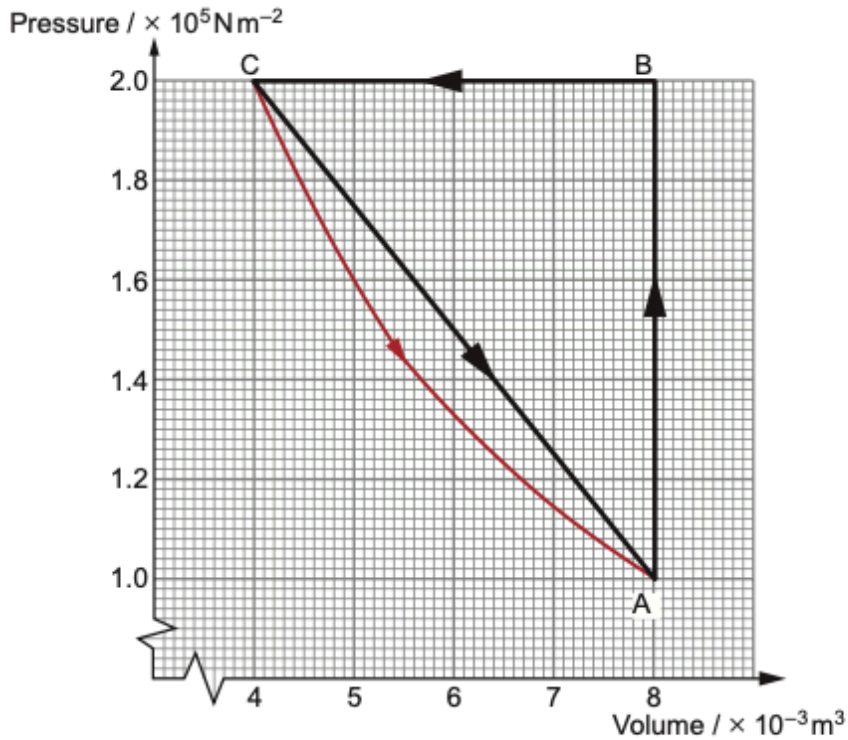
- (i) Complete the table by describing each process in terms of pressure and volume. The description for process $A \rightarrow B$ is already inserted. For each process state if any work is done, and if so indicate if it is done *on* or *by* the gas. *No calculations are required.* [5]

Process	Description of process	Work done on / by gas (if any)
$A \rightarrow B$	Increase in pressure at constant volume	
$B \rightarrow C$		
$C \rightarrow A$		

- (ii) Determine the net work done on the gas during the entire cycle.

[2]

- (c) The same mass of gas is taken around the cycle $A \rightarrow B \rightarrow C \rightarrow A$ for a second time. For this cycle the temperature is kept constant between C and A. The new path from C to A is shown in red on the graph below, together with the original path.



- (i) Show that the temperature is constant along the new path from C to A.

[3]

- (ii) Estimate the difference in the net work done in the two cycles.

[2]

Question			Marking details	Marks available				Maths	Prac												
				AO1	AO2	AO3	Total														
3	(a)		ΔU increase (or change) in internal energy of a system (1) Q heat flowing into the system (1) W work done by the system (1) Reference to system at least once otherwise award a maximum of 2 marks	3			3														
	(b)	(i)	<table border="1"><thead><tr><th>Process</th><th>Description of process</th><th>Work done on/by gas (if any)</th></tr></thead><tbody><tr><td>A $\rightarrow$ B</td><td>Increase in pressure at constant volume</td><td>No work done (1) – AO2</td></tr><tr><td>B $\rightarrow$ C</td><td>Decrease in volume at constant pressure (1) – AO1</td><td>Work done on gas (1) – AO2</td></tr><tr><td>C $\rightarrow$ A</td><td>[Linear] decrease in pressure with increasing volume (1) – AO1</td><td>Work done by gas (1) – AO2</td></tr></tbody></table>	Process	Description of process	Work done on/by gas (if any)	A $\rightarrow$ B	Increase in pressure at constant volume	No work done (1) – AO2	B $\rightarrow$ C	Decrease in volume at constant pressure (1) – AO1	Work done on gas (1) – AO2	C $\rightarrow$ A	[Linear] decrease in pressure with increasing volume (1) – AO1	Work done by gas (1) – AO2	2	3		5		
Process	Description of process	Work done on/by gas (if any)																			
A $\rightarrow$ B	Increase in pressure at constant volume	No work done (1) – AO2																			
B $\rightarrow$ C	Decrease in volume at constant pressure (1) – AO1	Work done on gas (1) – AO2																			
C $\rightarrow$ A	[Linear] decrease in pressure with increasing volume (1) – AO1	Work done by gas (1) – AO2																			
		(ii)	Work done on the gas is given by the "area" enclosed $= \frac{1}{2}(8 - 4)10^{-3}(2 - 1)10^5$ (1) $= \frac{1}{2}(4)(1)10^2 = 200$ [J] (1)		2		2	2													
	(c)	(i)	Use of $PV = nRT$ and note T constant (1) At A $PV = (1 \times 10^5)(8 \times 10^{-3}) = 800$ [J] (or at C $PV = (2 \times 10^5)(4 \times 10^{-3}) = 800$ [J]) (1) Check any point on curve e.g. <table border="1"><thead><tr><th>If V (m^3)</th><th>P (Pa)</th></tr></thead><tbody><tr><td>5.0×10^{-3}</td><td>1.60×10^5</td></tr><tr><td>6.0×10^{-3}</td><td>1.33×10^5</td></tr><tr><td>7.0×10^{-3}</td><td>1.14×10^5</td></tr></tbody></table> (1)	If V (m^3)	P (Pa)	5.0×10^{-3}	1.60×10^5	6.0×10^{-3}	1.33×10^5	7.0×10^{-3}	1.14×10^5	1	1		3	3					
If V (m^3)	P (Pa)																				
5.0×10^{-3}	1.60×10^5																				
6.0×10^{-3}	1.33×10^5																				
7.0×10^{-3}	1.14×10^5																				

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
		(ii)	Attempt at calculation of area – [accept small square count of between 170 and 280 i.e. $225 \pm \sim 25\%$] (1) If 225 i.e. 2.25 large squares, work difference = $2 \Delta V \Delta p$ (ΔV and Δp for large square) $= 2.25 (1 \times 10^{-3})(0.2 \times 10^5) = 45$ J with uncertainty of 25%, accept a value between 34 and 56 J provided method correct (1)			2	2	2	
			Question 3 total	6	7	2	15	7	0

2018

3. (a) Give the definition of the *specific heat capacity* of a material. [1]

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- (b) An insulating flask contains $0.6 \times 10^{-3} \text{ m}^3$ of water at 19.5°C . A volume $1.6 \times 10^{-3} \text{ m}^3$ of boiling water at 100.0°C is poured into the flask.

(specific heat capacity of water, $c = 4\,200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$; density of water, $\rho = 1\,000 \text{ kg m}^{-3}$)

- (i) Determine the final temperature of the water in the flask. [3]

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- (ii) Calculate the heat lost by the boiling water. [3]

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- (iii) Justify the statement that no work is done on/by the boiling water as it cools. No further calculations are needed. [1]

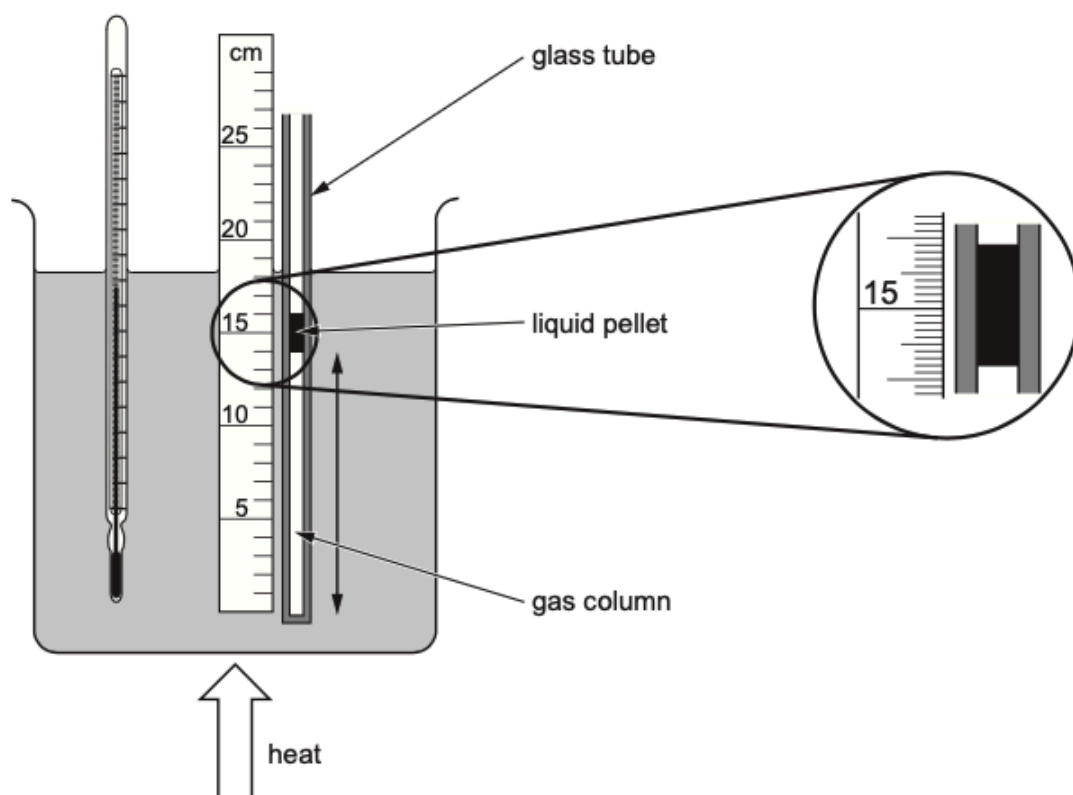
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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)		The specific heat capacity is the heat required to increase the temperature of 1 kg [or unit mass] of a material by 1 °C (1 degree K – or one degree/ unit temperature rise). [Alternative: equation with all terms fully defined.]	1			1		
	(b)	(i)	Idea: heat lost by boiling water=heat gained by other water (1) $1.6 \times 10^{-3} (100.0 - \theta_f) = 0.6 \times 10^{-3} (\theta_f - 19.5)$ (1) $\theta_f = 78.0$ °C (1) Alternative (for second mark) If $\Delta\theta$ is temperature rise of water in flask $1.6 \times 10^{-3} (80.5 - \Delta\theta) = 0.6 \times 10^{-3} \Delta\theta$ (✓)			3	3	2	
		(ii)	Mass of water = $1.6 \times 10^{-3} \times 1000 = 1.6$ [kg] (1) Application of $m = \rho V$ [even if $0.6 \times 10^{-3} \text{ m}^3$ used $\rightarrow 0.6$ kg] [Can be credited from (b)(i)] Heat lost = $1.6(100.0 - 78.0 \text{ ecf}) \times 4200$ (1) = $1.48 \times 10^5 \text{ J}$ (1) 148 J $\rightarrow$ 1 mark		3		3	3	
		(iii)	[Work done is negligible as] negligible / no change in volume.			1	1		
			Question 3 total	1	3	4	8	5	0

2019

1. A student performs an experiment to estimate the Celsius temperature of absolute zero.



The length, l , of the gas column and its temperature, θ , are measured at atmospheric pressure, 1.01×10^5 Pa. The temperature of the water is initially 0.0°C . It is then increased to 80.0°C . Readings are taken at 20.0°C intervals. The scale used to measure the length gives an uncertainty of ± 0.1 cm at the top of the column and an uncertainty of ± 0.1 cm at the bottom.

Values of θ and l are recorded in the table below.

$\theta / ^\circ\text{C}$	l / cm
0.0	11.5
20.0	12.5
40.0	13.2
60.0	14.2
80.0	15.0

(a) Justify the number of significant figures used to record the length, l . [2]

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(b) The glass tube in which the gas is trapped has an internal diameter of (1.5 ± 0.1) mm.

(i) Calculate the volume of the gas at 0.0°C . [2]

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(ii) Show that the percentage uncertainty in this volume, V , is approximately 15%. [3]

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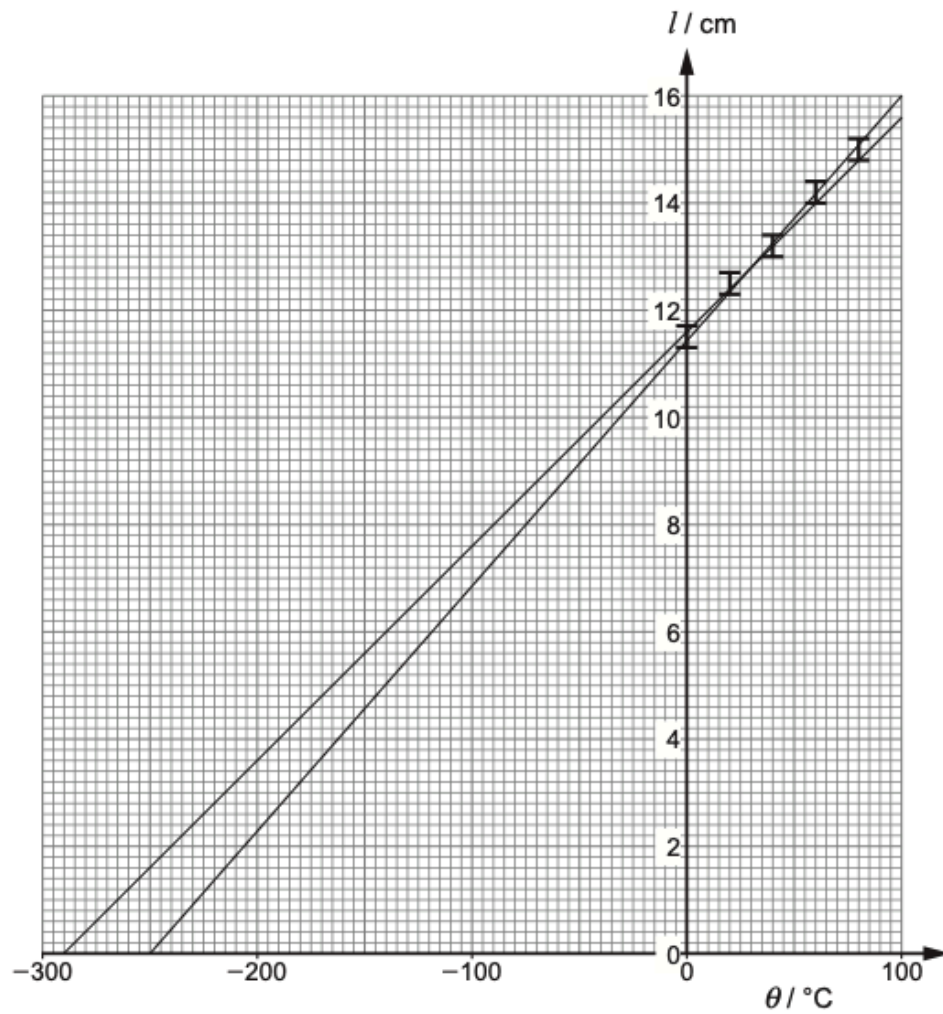
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- (c) The length, l , is plotted against the temperature, θ . Lines of maximum and minimum gradient consistent with the error bars have been drawn.



- (i) Use the graph to estimate the temperature of absolute zero in $^\circ\text{C}$, together with its **absolute** uncertainty. [2]

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(ii) Justify the statement that *the results are consistent with the ideal gas equation*. [4]

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(d) State what happens to the behaviour of the gas molecules as the temperature approaches absolute zero. [1]

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(e) The scale resolution of the ruler affects the final uncertainty in the value of absolute zero. Identify **two** other factors that could affect the accuracy. [2]

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		Award 1 mark for ± 0.1 [cm] or measure to resolution of ruler Award 2 marks for ± 0.2 [cm]	1	1		2	1	2
	(b)	(i)	$V = \pi \left(\frac{d}{2}\right)^2 l = \pi \left(\frac{1.5 \times 10^{-3}}{2}\right)^2 \times 11.5 \times 10^{-2}$ (subst and convincing change of units for l and d) (1) $= 2.03 \times 10^{-7} \text{ m}^3$ or 0.2 cm^3 or 203 mm^3 (1) unit mark	1	1		2	2	2
		(ii)	$p_v = 2p_d + p_i = 2 \frac{0.1}{1.5}(1) + \frac{0.2}{11.5}(1)$ (or by impl.) $= 0.133 + 0.017 = 0.150$ (1) allow ecf for this mark, on the use of 0.1 instead of 0.2 only Hence $p_v = 15\%$ (Alternative: use percentages throughout)	1	1 1		3	3	3
	(c)	(i)	From intercepts with x -axis, mean = -270 [$^{\circ}\text{C}$] (1) Uncertainty = 20 [$^{\circ}\text{C}$] (1) Award 1 mark only if 270 used		2		2	1	2
		(ii)	Any 4 $\times$(1) from: -Straight line -Intercept is consistent i.e. -273°C / line would go through the origin if temp plotted in K -Passes through all error bars -Volume linked to length - $V \propto T$ or $l \propto T$			4	4	1	4
	(d)		Kinetic or internal energy (or velocity) approaches minimum / zero. Accept very little KE / stopped moving / molecules stop / stops vibrating Don't accept KE decreases greatly / superconduct or superfluid	1			1		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(e)		Any 2 $\times$(1) from: -Meniscus or equivalent linked to liquid pellet -Moving readings -Expansion of glass -Gas not ideal -Variation in atmospheric pressure -Gas and liquid at different temperatures -Friction / viscosity of liquid pellet -Parallax / looking at eye level -Ruler not parallel to tube don't accept just ruler vertical Accept inaccuracy of thermometer Don't accept resolution of thermometer			2	2		2
			Question 1 total	4	6	6	16	8	15

2023

1. The first law of thermodynamics may be written as:

$$\Delta U = Q - W$$

- (a) State the meaning of: [3]

ΔU

Q

W

- (b) A closed cylinder fitted with a leak-proof piston as one end contains 0.060 mol of an ideal monatomic gas. The volume of the system is $1.4 \times 10^{-3} \text{ m}^3$ and the pressure of the gas is 100 kPa. Determine the temperature of the gas. [2]

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- (c) The gas is heated and expands at constant pressure to a volume of $2.0 \times 10^{-3} \text{ m}^3$. Determine:

- (i) the work done by the gas, [2]

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- (ii) the increase in the internal energy of the gas, [2]

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- (iii) the heat transferred to the gas. [1]

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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			ΔU : is the increase in the internal energy of a system (1) accept change Q : is the heat entering (1) the system W : is the work done <u>by</u> (1) the system "system" or "gas" or equivalent needs to be used once correctly for full marks.	3			3		
	(b)			Substitution into $PV = nRT$ (1) $T = \frac{PV}{nR} = \frac{(100 \times 10^3)(1.4 \times 10^{-3})}{(0.06)(8.31)} = 280.8 \text{ K unit mark (1)}$	1	1		2	2	
	(c)	(i)		Work done (by gas) = $P \Delta V = (100 \times 10^3)(2.0 - 1.4) \times 10^{-3} (1)$ = 60 [J] (1) Award 1 mark for 200 or 140 [J]	1	1		2	2	
		(ii)		Final temperature: $T = \frac{PV}{nR} = \frac{(100 \times 10^3)(2.0 \times 10^{-3})}{(0.06)(8.31)} = 401.1 \text{ [K] (1)}$ Increase in internal energy = $\frac{3}{2} nR \Delta T =$ $1.5(0.06)(8.31)(401.1 - 280.8 \text{ecf}) = 90.0 \text{ [J] (1)}$ Alternative: Use of $\frac{3}{2} P \Delta V$ (1) = 90 [J] (1)		2		2	2	
		(iii)		Heat transferred $Q = \Delta U + W = 90.0 \text{ ecf} + 60.0 \text{ ecf} = 150.0 \text{ [J]}$		1		1	1	
				Question 1 total	5	5	0	10	7	0

4. (a) Explain what is meant by specific heat capacity. [2]

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- (b) A volume of $1.2 \times 10^{-3} \text{ m}^3$ of water is heated in a 3 kW kettle.

Specific heat capacity of water (and tea) = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific heat capacity of milk = $3900 \text{ J kg}^{-1} \text{ K}^{-1}$

Density of water = $1.00 \times 10^3 \text{ kg m}^{-3}$

Density of milk = $1.03 \times 10^3 \text{ kg m}^{-3}$

- (i) Determine the time it takes to increase the temperature of the water from 18°C to 100°C . [3]

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- (ii) A volume of $0.25 \times 10^{-3} \text{ m}^3$ of the heated water is used to make tea and this is poured into a thermos flask. The temperature of the tea when in the flask is 95°C . Milk at 5°C is then poured into the flask. Determine whether $3.6 \times 10^{-5} \text{ m}^3$ of milk will lower the temperature of the mixture to the required temperature of 84°C . [4]

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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			The {heat / energy} required to {raise / change} [the temperature of] 1 kg (1) by 1 °C or K (1)	2			2		
	(b)	(i)		Mass of the water = density $\times$ volume = $10^3 \times (1.2 \times 10^{-3}) = 1.2$ kg (1) Heat required to increase temperature = $shc \times \text{mass} \times \text{increase in temperature} = 4200 \times 1.2 \times (100 - 18) = [413\,280 \text{ J}]$ (1) Time required = $\frac{\text{heat required}}{\text{power}} = \frac{413\,280}{3 \times 10^3} = 137.8$ [s] (1) Or 2 min 18 s		3		3	3	
		(ii)		Use of density equation on milk [expect 0.037 kg] (1) Attempt at conservation of energy (1) e.g. Heat lost by the tea = Heat gained by the milk [expect 11 424 and 11 550 [J]] Correct equation set up (1) e.g. $(95 - 84)(4200)(0.25) = (84 - 5)(3900)m_{\text{milk}}$ or $(95 - \theta)(4200)(0.25) = (\theta - 5)(3900)m_{\text{milk}}$ or equivalent equation set up to calculate one of the variables or $0.25 \times 4200 \times 95 + 0.03708 \times 3900 \times 5 = (0.25 \times 4200 + 0.03708 \times 3900) \times T$ (note that this line also works if temperature is in K) Final answer and comment (1) e.g. $V = 3.64 \times 10^{-5} \text{ m}^3$ and so about right or final temperature = 84.1 °C and so about right			4	4	3	
				Question 4 total	2	3	4	9	6	0

Nuclear Decay

2017

1. (a) Describe an experiment that uses various absorbers to identify the type(s) of radiations emitted by a radioactive sample. Justify all steps in the experiment. [4]

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- (b) A radioactive sample of material has a half-life of 11.4 days and an initial activity of A_0 . Determine:

- (i) the decay constant; [2]

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(ii) the activity of the sample after 57.0 days in terms of A_0 ; [2]

(iii) the percentage decrease in the number of nuclei in the sample after 57.0 days. [3]

Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
1	(a)	Diagram showing source, absorber and detector or equivalent stated in words (1) I. Take measurement with no source and no paper/aluminium – [to measure the background radiation] or show awareness of background. II. Measure count rate with no paper/aluminium. (1 for both I and II) III. Insert [a few sheets of] paper between the source and receiver and take measurement. If reduction from count in II then alpha particles present. (1) IV. Insert [a few mm of] aluminium between the source and receiver and take measurement. If reduction in the count from that in III then beta particles are also present and if count is still above the background level, then gamma radiation is present – this penetrates the aluminium. (1)	4			4		4
	(b) (i)	$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{(11.4 \times 24 \times 60 \times 60)} \text{ Substitution (1)}$ $7.037 \times 10^{-7} \text{ s}^{-1} \text{ unit (1)}$ Alternative solution: $\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{11.4} \text{ substitution (1) } = 0.0608 \text{ days}^{-1} \text{ unit (1)}$	1	1		2	2	
	(ii)	$\frac{57.0}{11.4} = 5 \text{ so 5 half lives (1)}$ $\text{Activity} = \frac{1}{2^5} A_0 = \frac{1}{32} A_0 \text{ (1)}$ Alternative solution: $A = A_0 e^{-\lambda t}; t = 5T_{\frac{1}{2}} \text{ and } \lambda = \frac{\ln 2}{T_{\frac{1}{2}}}; \text{ substitution } A = A_0 e^{-5 \ln 2} (1)$ $= A_0 \frac{1}{2^5} = \frac{1}{32} A_0 (1)$	1	1		2	2	

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
		(iii)	Let the initial number of particles be N_0, so number of particles remaining after 57 days = $\frac{1}{32} N_0$ i.e. $A \propto N(1)$ Decrease in the number of nuclei in 57 days = $\left(1 - \frac{1}{32}\right) N_0$ i.e. decrease (1) Percentage decrease = $\frac{\left(1 - \frac{1}{32}\right) N_0}{N_0} \times 100\% = 96.875\%$ i.e. percentage (1) Answer only of 3% award 2 marks only	1	1	1	3	3	
			Question 1 total	7	4	0	11	7	4

2018

5. (a) Radon gas decays by emitting α -particles. It has a half-life of 3.8 days. Calculate the **percentage reduction** in the activity of a sample of radon after 12 days. [4]

- (b) A student makes the following measurements for a radioactive source using the indicated absorber between the source and detector.

Absorber	Counts per minute
none	1 004
sheet of paper	597
2 mm of aluminium	23
15 cm of lead	27

Explain these observations. [4]

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)		$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{3.8} \text{ day}^{-1} [=0.182 \text{ day}^{-1}] (1)$ Activity after 12 days $A = A_0 \exp\left(-\frac{12 \ln 2}{3.8}\right) (1)$ substitution $= 0.112 A_0 (1)$ $\therefore$ % reduction = 88.8%(1) (Accept variation because of rounding) Alternative Number of half-lives, $n = \frac{12}{3.8} = 3.16$ [or by implication] (1) Fraction after 12 days $= 2^{-3.16} (1) = 0.112 (1) = 11.2\%$ $\therefore$ Percentage reduction = 88.8% (1)		4		4	4	

	(b)		Any 4 $\times$ (1) 1) Counts is reduced significantly (or equivalent alternatives e.g. by almost a half) by the paper, so alpha particles present (✓) 2) Another significant reduction (or alternative e.g. essentially all of the remaining radiation is stopped) by the aluminium, so beta particles present (✓) 3) Count with lead is larger than with aluminium (but almost the same) so no gamma present (✓) 4) The measured counts with aluminium and lead are essentially the same, so this is because of background radiation / the background radiation is approximately 25 counts per minute. (✓) 5) Randomness of nuclear decay is the reason for increased value with lead. (✓)		4		4		4
			Question 5 total	0	8	0	8	4	4

2022

1. (a) Radon is a radioactive gas and emits alpha particles.

(i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]

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(ii) The half-life of radon is 3.3×10^5 s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]

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(iii) Discuss the extent to which alpha particles are a risk to the human body. [3]

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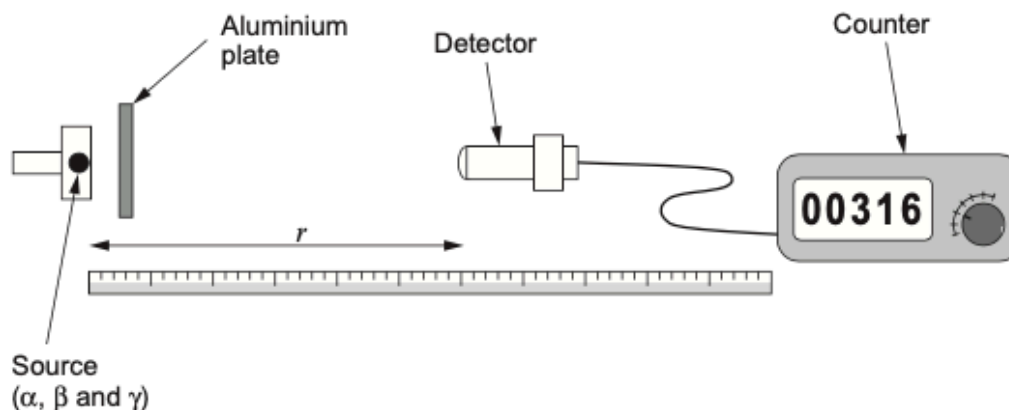
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	[The <i>activity</i> is the] rate of decay or [the number of] disintegrations / decays per second (or unit time [of a sample of radioactive nuclei] (1) Accept number of radioactive emissions per second Becquerel or becquerel or Bq (1) Accept s ⁻¹	2			2		
		(ii)	$\lambda = \frac{\ln 2}{3.3 \times 10^5} = 2.1 \times 10^{-6} \text{ s}^{-1}$ (1) $e^{\lambda t} = \frac{A_0}{A}; \quad t = \frac{1}{\lambda} \ln \left(\frac{A_0}{A} \right)$ (1) $t = \frac{1}{(2.1 \times 10^{-6})} \ln \left(\frac{1}{0.2} \right) = 7.7 \times 10^5 \text{ s} = [8.9 \text{ day}]$ (1) Alternative: Taking logs of $A = \frac{A_0}{2^n}$ e.g. $\ln A = \ln A_0 - n \ln 2$ or $n \ln 2 = \ln \frac{A_0}{A}$ (1) Substitution: $n \ln 2 = \ln \frac{100}{20}$ [leads to $n = 2.32$] (1) Correct answer $7.7 \times 10^5 \text{ s} = [8.9 \text{ day}]$ (1)		3		3	3	
		(iii)	Safe outside the body or stopped by the skin or danger or dangerous if swallowed / inhaled (1) <u>Highly</u> ionising (1) Any reference to damage e.g. damage DNA, kills cells, cause cancer, cause radiation poisoning, causes mutations (1)			3	3		

5. A student says that the radiation intensity, I , of gamma rays increases with decreasing distance, r , from a source such that:

$$I \propto \frac{1}{r^n}$$

She wishes to determine the value of n .

The student uses a source of alpha, beta and gamma radiation. She measures the intensity of the rays at set distances from the source by determining the count over intervals of one minute. She also takes measurements without the source at the start and at the end of the experiment.



- (a) (i) Suggest why a thin aluminium plate is placed near the source. [1]

- (ii) Explain why readings are taken without the source, at the start **and** at the end of the experiment. [2]

- (b) At each distance, r , the counts were measured over one minute intervals three times. The uncertainty in r is ± 0.005 m. The measurements are recorded in the table.

Complete the table.
Space for calculations.

[2]

Distance, r / m	Count				Uncertainty in mean count
	First reading	Second reading	Third reading	Mean	
without source	19	22	25	22	3
1.000	110	130	136	125	13
0.800	195	165	178		
0.600	270	316	300		
0.400	661	604	651	639	29
without source	20	20	25	22	3

- (c) (i) The student subtracts the mean count without the source from the mean count for each distance to obtain the corrected mean count, N . She also determines the uncertainty in each of these corrected mean counts.

Complete the table.
Space for calculations.

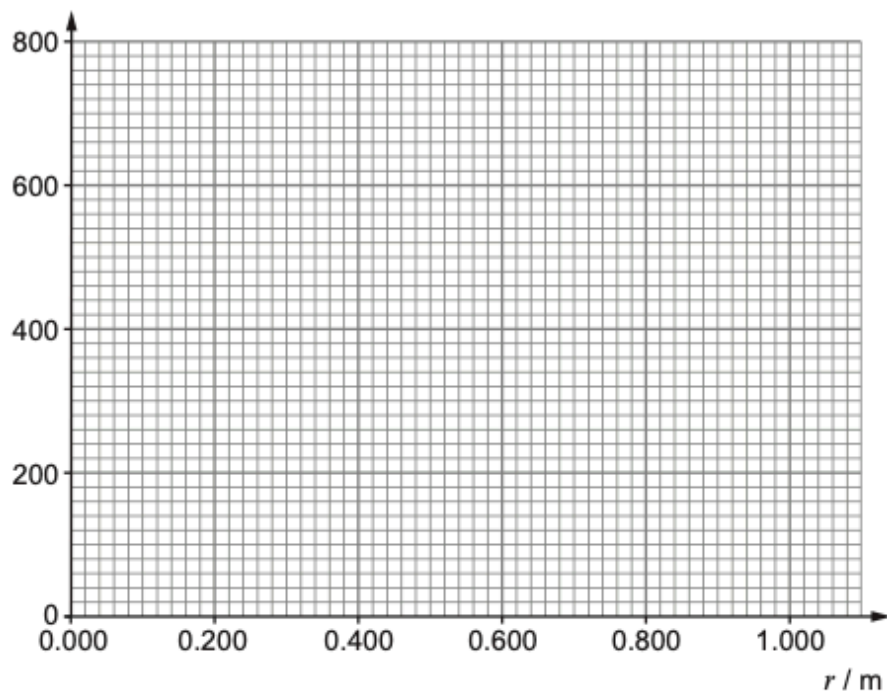
[2]

Distance, r / m	Corrected mean count, N	Uncertainty in corrected mean count
1.000	103	16
0.800		
0.600		
0.400		

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- (ii) Plot a graph of the corrected mean count, N , against distance, r . **Error bars are not required.** [3]

Corrected mean count, N



- (iii) The student claims that error bars may be shown on the graph for the corrected mean count, but not the distance. Justify her claim. Calculations are not required. [1]

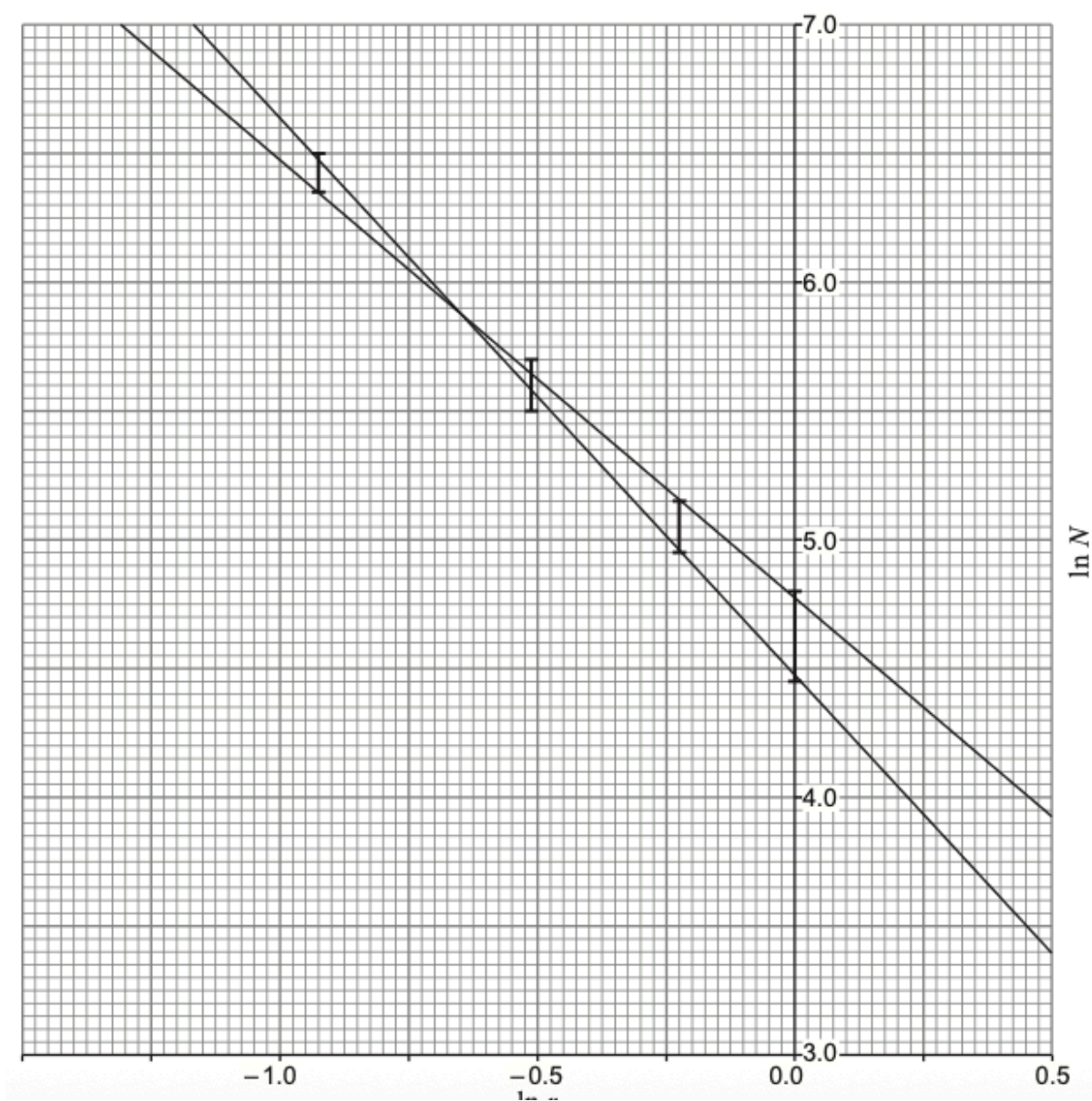
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- (d) Her friend suggests using a logarithmic graph. She calculates the logarithmic values in the table below.

Distance, r / m	$\ln r$	$\ln N$
1.000	0.00	4.63
0.800	-0.22	5.06
0.600	-0.51	5.61
0.400	-0.92	6.42

The data are plotted on the graph.

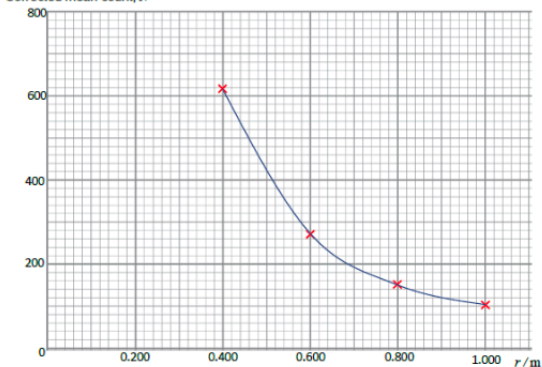


For a distance of $r = 1.000\text{ m}$ use the table in part (c) to show that the uncertainty in the logarithmic counts for this distance is approximately 0.15. [3]

- (e) (i) Starting with intensity: $I \propto \frac{1}{r^n}$ justify how the logarithmic graph can be used to determine n . [3]

- (ii) Use the maximum and minimum gradients to determine n and the **absolute** uncertainty in its value. [4]

Question			Marking details	Marks available																																																			
				AO1	AO2	AO3	Total	Maths	Prac																																														
5	(a)	(i)	Absorb alpha and beta or only allow gamma through. Accept absorb beta only [due to 40 cm gap]			1	1		1																																														
		(ii)	Background radiation (1) To check for consistency / check for variations (1) Don't accept for a mean			2	2		2																																														
	(b)		<table><tr><th rowspan="2">Distance, r / m</th><th colspan="4">Count</th><th rowspan="2">Uncertainty in mean count</th></tr><tr><th>First reading</th><th>Second reading</th><th>Third reading</th><th>Mean count</th></tr><tr><td>Without source</td><td>19</td><td>22</td><td>25</td><td>22</td><td>3</td></tr><tr><td>1.000</td><td>110</td><td>130</td><td>136</td><td>125</td><td>13</td></tr><tr><td>0.800</td><td>195</td><td>165</td><td>178</td><td>179</td><td>15 or 14 or 16</td></tr><tr><td>0.600</td><td>270</td><td>316</td><td>300</td><td>295</td><td>23 or 21 or 22 or 24 or 25</td></tr><tr><td>0.400</td><td>661</td><td>604</td><td>651</td><td>639</td><td>29</td></tr><tr><td>Without source</td><td>20</td><td>20</td><td>25</td><td>22</td><td>3</td></tr></table> 1 mark per column.	Distance, r / m	Count				Uncertainty in mean count	First reading	Second reading	Third reading	Mean count	Without source	19	22	25	22	3	1.000	110	130	136	125	13	0.800	195	165	178	179	15 or 14 or 16	0.600	270	316	300	295	23 or 21 or 22 or 24 or 25	0.400	661	604	651	639	29	Without source	20	20	25	22	3		2		2	2	2
Distance, r / m	Count				Uncertainty in mean count																																																		
	First reading	Second reading	Third reading	Mean count																																																			
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Without source	20	20	25	22	3																																																		
	(c)	(i)	<table><tr><th>Distance, r / m</th><th>Corrected mean count, N</th><th>Uncertainty in corrected mean count</th></tr><tr><td>1.000</td><td>103</td><td>16</td></tr><tr><td>0.800</td><td>157</td><td>18</td></tr><tr><td>0.600</td><td>273</td><td>26</td></tr><tr><td>0.400</td><td>617</td><td>32</td></tr></table> 1 mark per column ecf on previous table	Distance, r / m	Corrected mean count, N	Uncertainty in corrected mean count	1.000	103	16	0.800	157	18	0.600	273	26	0.400	617	32		2		2	2	2																															
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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	 All 4 points correct (2) ecf 3 points correct (1) One smooth curve passing through all points $\pm \frac{1}{2}$ small square (1)		3		3	3	3
		(iii)	Error bars in the corrected mean count are large enough to be shown on the graph [approx. 2 units min], but the uncertainty in r is too small [less than one unit]			1	1		1

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
	(d)	$\ln 119 = 4.78$ or $\ln 87 = 4.47$ (1) $\ln 103 = 4.63$ or the other extreme (1) $4.78 - 4.63$ or $4.66 - 4.47 = [\text{approx. } 0.15]$ or $\frac{0.31}{2}$ (1) No ecf allowed (since 103 and 16 are given) Award 1 mark for $\frac{16}{103} = 0.15, 0.155, 0.16$ Award 2 marks for statement absolute uncertainty in log = fractional uncertainty in count		3		3	3	3
	(e) (i)	Taking logs: $\ln(N) = -n \ln(r) + \ln(k)$ (1) (accept $\ln I$ in lieu of $\ln N$) Plotting $\ln(N)$ on the y-axis against $\ln(r)$ on the x-axis or comparison with $y = mx + c$ (1) Gradient = $-n$ (1)			3	3	3	3
	(ii)	Minimum gradient = $\frac{3.94-7.00}{0.50-(-1.30)} = [-]1.69 \pm 0.03$ (1) Maximum gradient = $\frac{3.40-7.00}{0.50-(-1.17)} = [-]2.16 \pm 0.03$ (1) Mean gradient = $[-]1.93 \pm 0.03$ (1) ecf Uncertainty = $\frac{1}{2} (-1.69 - (-2.16)) = 0.20 - 0.28$ (1) ecf accept 0.2 No sig fig penalty		4		4	4	4
		Question 5 total	0	14	7	21	17	21

2023

3. (a) State what is meant by alpha emission. [2]

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- (b) Americium-241 has a half-life of 432 years. Determine the percentage **decrease** in the activity of the element after 30 years. [4]

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- (c) A builder is asked to install a smoke detector. He researches smoke detectors and finds that the radioactive source in the detector emits alpha particles. He decides that it is too dangerous to install the detector. Discuss whether or not he has used science well in his decision making. [2]

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- by using different absorbers
- by using an electric field **or** a magnetic field.

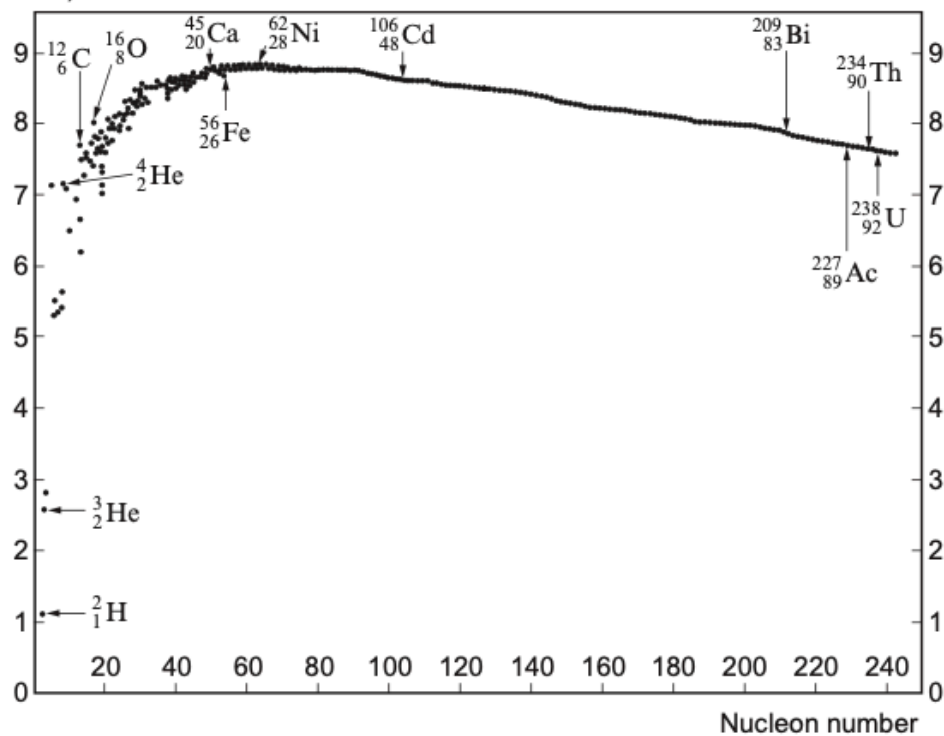
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Helium nuclei / 2 protons and 2 neutrons (1) accept ${}^4_2\text{He}$ Emitted from the <u>nucleus</u> [at high speeds] (1)	2			2		
	(b)			$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{432} = 1.60 \times 10^{-3} \text{ yr}^{-1}$ or $5.08 \times 10^{-11} \text{ s}^{-1}$ (1) Either: Substitution: $A = A_0 \exp(-(1.60 \times 10^{-3})(30))$ (1) $\frac{A}{A_0} 100\% = 95.3\%$ (1) % decrease in activity = $100 - 95.3 = 4.7\%$ (1) or: 30 years $\ll$ $\frac{1}{2}$ -life [or by impl.] (1) $\therefore$ Fractional decay in 30 years = $30\lambda = 0.048$ (1) $\therefore$ % decrease in activity = 4.8% (1) Accept 5% Alternative: 30 years is $30/432 = 0.07$ half-lives (1) $A = \frac{1}{2}^{0.07}$ (1) $= 0.95$ (1) So activity has decreased by 5% (1)	1	1		4	3	
	(c)			Any 2 $\times$ (1) from: - Alpha particles pose little risk unless inhaled or ingested. - Very few (or none) make it to the outside of the detector. - If they do, they are stopped by a few cm of air and are unable to penetrate surface of skin. To award both marks there must be a conclusion present i.e. a minimum of no stated.			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			Indicative content: Absorbers: Explain how absorbers can be used to distinguish between the three radiations – describe set up. Diagram or a clear explanation. Sheet of paper stops alpha. Few mm of aluminium will stop alpha and beta. Gamma penetrates both the paper and aluminium. Reference to background radiation. Magnetic field / Electric field: Explain how a magnetic / electric field can be used to distinguish between the three radiations – describe set up. Diagram and clear explanation. Deflection of alpha particles much smaller than for beta, charge positive. Deflection of beta particles opposite to that of alpha, as charge-negative. No deflection of gamma. Correct direction of alpha or beta in magnetic / electric field. 5-6 marks Absorbers and magnetic / electric field both covered comprehensively. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Either absorbers or magnetic / electric field covered comprehensively or limited account of both areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	4	2		6		6

Nuclear Energy

2017

- [4]



(b) The fusion of hydrogen to helium in the Sun may be represented by:



Particle	Mass / u
${}^1_1\text{H}$	1.00728
${}^4_2\text{He}$	4.00151
e^-	0.00055
ν_{e}	0.00000

$$1\text{ u} = 931\text{ MeV}$$

Calculate the energy released in this reaction in MeV.

[3]

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(c) Nuclear power has played a role in the generation of electricity in Wales. Wylfa Newydd on Anglesey may feature in further developments. Indicate a benefit and an issue that may arise from such projects and discuss their relative importance. [3]

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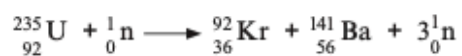
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Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
2	(a)	The most stable nuclei (or reference to elements near to peak i.e. Fe, Ni Ca) are therefore where the curve (or binding energy per nucleon) reaches its maximum. (1) Nuclei of small atomic mass number (lhs of graph) can combine to produce species of larger atomic mass number, [hence larger binding energy per nucleon. Energy is released]. Fusion. (1) Nuclei of large atomic mass number (on rhs of graph) break down to produce species of smaller atomic mass number, [hence larger binding energy per nucleon. Energy is released]. Fission. (1) Reference anywhere to there being energy released, when a <u>reduction in mass</u> occurs i.e. mass converted to energy. (1)	4			4		
	(b)	Mass defect = $4(1.00728) + 2(0.00055)$ (1) $- 4.00151 = [0.02871 \text{ u}]$ (1) Energy: $0.02871 \times 931 = 26.7 [\text{MeV}]$ (1)		3		3	3	
	(c)	Benefit: routine supply of energy (i.e. does not depend on weather) or jobs or no CO₂ emission (1) Issue: needs secure storage of radioactive waste (products) over a extended time period or reference to long half-lives or long build time or building extra transmission power lines (1) Reasoned conclusion (1)			3	3		
		Question 2 total	4	3	3	10	3	0

2018

4. (a) A fission nuclear reaction for uranium is:



The masses of the nuclei are:

$$\text{mass of } {}_{92}^{235}\text{U} = 235.01 \text{ u}$$

$$\text{mass of } {}_{36}^{92}\text{Kr} = 91.90 \text{ u}$$

$$\text{mass of } {}_{56}^{141}\text{Ba} = 140.89 \text{ u}$$

$$\text{mass of } {}_0^1\text{n} = 1.01 \text{ u}$$

$$1 \text{ u} = 931 \text{ MeV}$$

Calculate the energy released in this nuclear reaction.

[2]

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- (b) Explain carefully the relevance of the **binding energy per nucleon** curve to nuclear fission and fusion. [6 QER]

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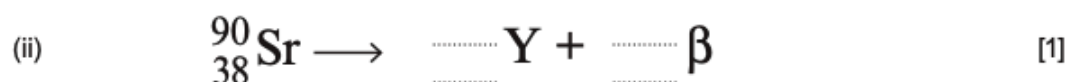
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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
4	(a)		Mass defect = $((235.01 + 1.01) - (91.90 + 140.89 + 3(1.01)))u$ [or $(235.01 - (91.90 + 140.89 + 2 \times 1.01))] = 0.20 u$ (1) Energy released = $0.20 \times 931 = 186 \text{ MeV}$ [190 MeV] (1) (Accept $3.0 \times 10^{-11} \text{ J}$) UNIT		2		2	2	
	(b)		General (G) G1: General shape of curve [correct asymmetry] & axes labelled G2: Maximum around nucleon number 60 / iron / nickel G3: Stability linked to BE/nuc to stability G4: Nucleons react to move towards maximum (on curve) G5: In doing so there is a loss / reduction in mass G6: Mass loss linked to energy released = $\Delta m c^2$ Fusion (Fu) Fu1: Smaller nucleon number nuclei combine Fu3: Larger nucleons of larger nucleon number formed: energy released, BE/nuc increases, stability increases, lower mass Fission (Fi) Fi1: Larger nucleon number split Fi3: Smaller nucleons of smaller nucleon number formed energy released, BE/nuc increases, stability increases, lower mass Additional point: gradient of curve larger in fusion region than fission resulting in more energy release (per nucleon) 5-6 marks At least 6 G points (6-8 G) 8 – 11 points <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i>	6			6		

2019

6. (a) Complete the following decay equations.



(b) Calculate the binding energy per nucleon for ${}_{38}^{90}\text{Sr}$. [4]

$$m_{\text{proton}} = 1.007\,276\,\text{u}$$

$$m_{\text{neutron}} = 1.008\,664\,\text{u}$$

$$m_{\text{electron}} = 0.000\,549\,\text{u}$$

$$\text{atomic mass of } {}_{38}^{90}\text{Sr} = 89.907\,738\,\text{u}$$

$$1\,\text{u} = 931\,\text{MeV}$$

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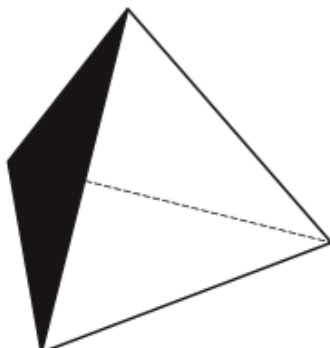
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- (c) Radioactive decay may be investigated in the laboratory using a dice analogy.

A student has 564 identical wooden dice. These dice are tetrahedrons, with one face painted black. A tetrahedron is a solid with four identical faces, each face being an equilateral triangle.



- (i) If the student throws one of the dice, what is the probability of the tetrahedron landing on the black face? [1]

- (ii) The student then throws all dice and counts and discards those that landed on the black face. The remaining dice are thrown again and the process is repeated multiple times. The number of dice discarded after each throw is recorded in the table, and the number remaining after each throw is calculated. The results are shown in the table, which also gives the fraction of dice remaining, where:

$$\text{Fraction of dice remaining after a throw} = \frac{\text{number of dice remaining after a throw, } R}{\text{number of dice thrown in that throw, } T}$$

Throw number, n	Number of dice thrown, T	Number of dice discarded after throw	Number of dice remaining after throw, R	Fraction remaining after throw, $\frac{R}{T}$
1	564	138	426	0.76
2	426	116	310	0.73
3	310	87	223	0.72
4	223	52	171	0.77
5	171	39	132	0.77
6	132	34	98	0.74
7	98	10	88	0.90
8	88	27	61	0.69
9	61	18	43	0.70
10	43	8	35	0.81

I. Explain why the expected fraction remaining after n throws is $(0.75)^n$. [2]

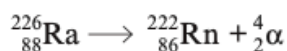
II. Hence predict the expected number remaining after 10 throws. [1]

III. Discuss to what extent the observations agree with those expected from theory. [3]

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			${}_{90}^{228}\text{Th} \rightarrow {}_{88}^{224}\text{Ra} + {}_2^4\alpha$ (1) ${}_{38}^{90}\text{Co} \rightarrow {}_{39}^{90}\text{Y} + {}_{-1}^0\beta$ (1)		2		2		
	(b)			Nucleon mass = 90.727 u or nucleon mass +38 e (90.747u) (1) Mass defect attempted with or without electrons 0.84 u or 0.82 u (1) × 931 and division by 90 (1) Answer = 8.69 [MeV per nucleon] (1) If electrons not taken into account answer = 8.47 [MeV per nucleon] award 3 marks 782 or 762 [MeV per nucleon] award 2 marks	1	1 1 1		4	3	
	(c)	(i)		Probability of landing on black face = $\frac{1}{4}$ or 0.25 or 25 %		1		1	1	1
		(ii)	I.	Probability of not decaying (i.e. of remaining) after 1 throw = 1 – 0.25 = 0.75 (1) Probability of remaining after 2 throws = 0.75^2 or probability of remaining after n throws = $(0.75)^n$ (1)		2		2	2	2
			II.	Number predicted = $N_0 \times (0.75)^n = 31.76 = 32$ Accept 31 or 31.76		1		1	1	1
			III.	Close to 0.75 for many throws or mean close to 0.75 or 32 is close to 35 or fits quite well with $(0.75)^n$ (1) Some further out e.g. 0.90 (1) Random process [these results are to be expected] (1)			3	3		3
				Question 6 total	1	9	3	13	7	7

2022

(b) Radium-226 decays into radon.



Assuming that 98% of the energy released in this process is converted to the kinetic energy of the alpha particle, use the data below to determine the speed of the alpha particle. [5]

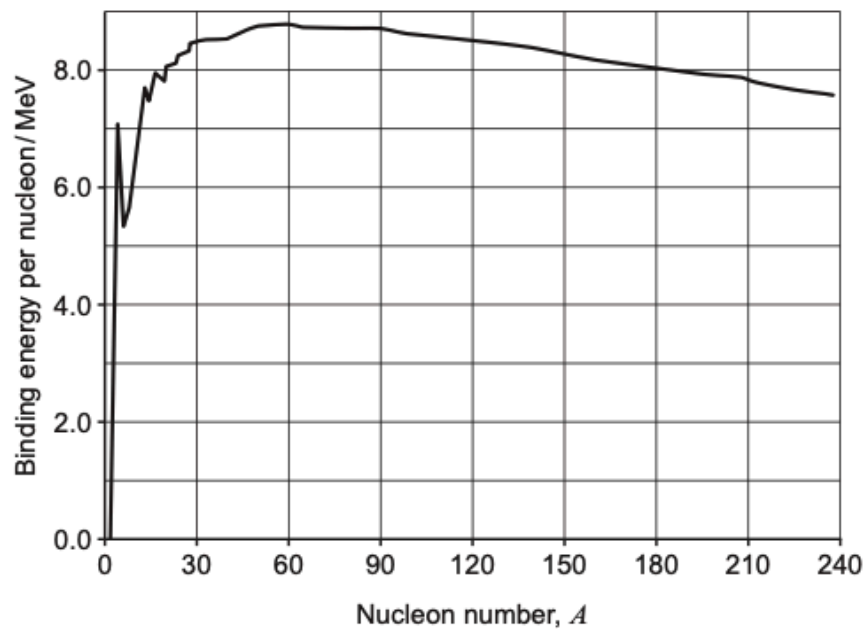
Atomic masses: ${}^{226}_{88}\text{Ra} = 226.025410\text{ u}$ ${}^{222}_{86}\text{Rn} = 222.017578\text{ u}$

$${}^4_2\text{He} = 4.002603 \text{ u} \quad m_{\text{electron}} = 0.000549 \text{ u}$$
$$1\text{ u} = 931\text{ MeV}$$

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		Attempt at masses: LHS – RHS i.e. $0.005\,229\,[\text{u}]$ or $0.004131\,[\text{u}]$ or $0.006327\,[\text{u}]$ (1) Attempt at conversion to MeV or mass i.e. $\times 931$ or $\times 1.66 \times 10^{-27}$ (1) Attempt at conversion to J i.e. $\times 1.6 \times 10^{-13}$ or $\times c^2$ (1) Expect $7.8 \times 10^{-13}\,[\text{J}]$ Rearrangement and incorporating 98 % or statement ignoring 98 % i.e. $v = \sqrt{\frac{2E}{m}}$ (1) N.B. Accept 4 u or 4.0026 u or 4.0015 u 1.52×10^7 or 1.35×10^7 or $1.67 \times 10^7\,[\text{ms}^{-1}]$ ecf on 98 % and one arithmetic slip (1)	1 1	1		5	5	
			Question 1 total	4	6	3	13	8	0

2023

5. (a) Nuclear energy can be released by fusion and fission. Discuss these processes in terms of energy and the stability of the nuclei. Use the graph of the binding energy per nucleon in your answer. [4]



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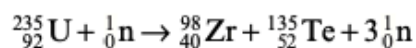
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- (b) One of the many possible fission reactions of uranium-235 is:



The masses are:

${}_{92}^{235}\text{U}$: 235.04393 u; ${}_{40}^{98}\text{Zr}$: 97.91273 u; ${}_{52}^{135}\text{Te}$: 134.91645 u, ${}_0^1\text{n}$: 1.00866 u
and 1 u = 931 MeV.

- (i) State what is meant by the Avogadro constant.

[1]

- (ii) Calculate the energy released by the fission of 0.0010 kg of uranium-235 in this reaction. Give your answer in J.

[5]

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Fusion definition (nuclei join to form larger nucleus and energy release) (1) Fission definition (larger nucleus splits to smaller nuclei and energy release) (1) N.B. energy release only required somewhere once Products are (usually) more stable / have higher BE/N (1) Low A numbers do fusion, high A numbers do fission (might be in diagram with arrows) (1)	4			4		
	(b)	(i)		The Avogadro constant is the {number of / 6.02×10^{23} } {particles / atoms / molecules} in 1 mole of a substance or in 12 g of ${}^{12}_6\text{C}$	1			1		
		(ii)		LHS - RHS i.e. $(235.04393 + 1.00866) - (97.91273 + 134.91645 + 3 \times 1.00866)$ $= 236.05259 - 235.85516 = 0.19743$ [u] (1) $\times 931$ or 1.66×10^{-27} (1) Convert from eV into J or kg into J i.e. 2.94×10^{-11} [J] (1) Method for number of atoms e.g. $\frac{(6.02 \times 10^{23})}{235} 1 = 2.56 \times 10^{21}$ or $\frac{0.001}{235 \times 1.66 \times 10^{-27}}$ (1) Energy released = 7.5×10^{10} [J] (1)		5		5	5	
				Question 5 total	5	5	0	10	5	0

Section B

2017

SECTION B

Answer all questions.

Read through the following article carefully.

Lasers in Outer Space

Paragraph

Einstein's theory of stimulated emission, published in 1916, laid the foundation for lasers but it took humans another 44 years before the first successful ruby laser was produced. However, seven years beforehand a maser had been produced. This is similar to a laser but involves microwaves instead of light – microwave amplification by stimulated emission of radiation.

1

It turns out that masers are easier to build than lasers because the lifetime of metastable energy levels tends to be proportional to $\frac{1}{\text{frequency of radiation}}$.

2

The ingenuity of humankind is often marvelled upon when such complicated devices as lasers and masers are used in devices such as DVD players and atomic clocks. However, nature itself is somewhat more modest – natural masers have been produced for billions of years in the atmospheres of stars, comets, star-forming regions, supernova remnants and even super massive black holes. And all this time the Universe has kept quiet about its technological achievements.

3

Essentially what happens in star masers is this – light and explosions from stars excite nearby gas regions. In these nearby gases, light and collisions lead to high energy levels becoming populated by electrons. Some of these higher energy levels will be metastable, setting up a maser amplifying region. As in any laser, the process has to start with a spontaneous emission accidentally shooting off in the correct direction, but afterwards, stimulated emission takes over and the maser beam starts towards infinity; infinity is stretching the truth a bit but it does sound good. In reality the intensity of the maser beam increases rapidly and the beam propagates at the speed of light as em radiation tends to do.

4

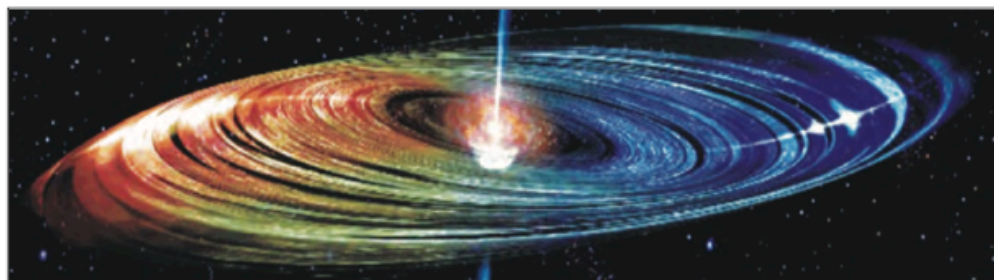


Diagram 1

In all fairness to humans, there is one aspect of laser design that nature has not succeeded in producing by itself and that is the resonant cavity of a laser. Multiple passes through a laser amplifying medium improves the quality of laser operation immensely. This is achieved among humans using two mirrors but as yet this technological advancement seems to be missing in gas regions around stars.

5

Although a resonant cavity seems to be missing in natural masers there does seem to be a high degree of beaming. Small differences across the irregularly shaped maser cloud disk lead to vast differences in intensity due to exponential gain. The directions in the gas disk that have a longer length of population inversion will appear much brighter (as the increased maser amplification leads to an exponential increase). The majority of the radiation will emerge along this line of greatest length in a "beam"; this is termed *beaming*. Not quite as good as a laboratory laser cavity but pretty impressive for a seemingly random gas cloud.

Megamasers is the term used for water masers in the gas cloud around black holes.

Most large galaxies with a nucleus and a bulge have, at their centres, a supermassive black hole. When the black hole is actively accumulating matter, it releases a tremendous amount of energy, and the galaxy is said to have an active nucleus. Buffered by the dust in the flow of matter into the black hole, molecules can survive there and get energised by collisions with other molecules and dust particles. Water molecules are common in this environment, and they can emit maser radiation. A water maser can shine a beam of microwaves at a very specific frequency, 22.235 GHz, which corresponds to microwaves about 1 cm in wavelength. The primary reason megamasers have been studied in detail is that they make excellent tools to help us understand the environment around black holes. Small changes in the detected frequency of the masers, owing to the Doppler effect, mean that we can determine the line-of-sight velocity of maser clouds very precisely. All this means that the mass of black holes can be determined 20 times more accurately.

Another primary science goal that we can address with studies of water megamasers is measuring distances to galaxies. Measuring distances is a notoriously difficult problem in astronomy, and one of the most important. By analysing the internal dynamics of water maser systems, we can measure the rotation velocity, v , of maser clouds as they orbit the black hole from the Doppler shift of the maser lines. We can also measure the centripetal acceleration, a , of maser clouds by observing how the Doppler shift velocity changes over time. Using the simple relation for centripetal acceleration:

$$a = \frac{v^2}{r}$$

we can then calculate the radius of the gas 'disk' (see below) orbiting the black hole.

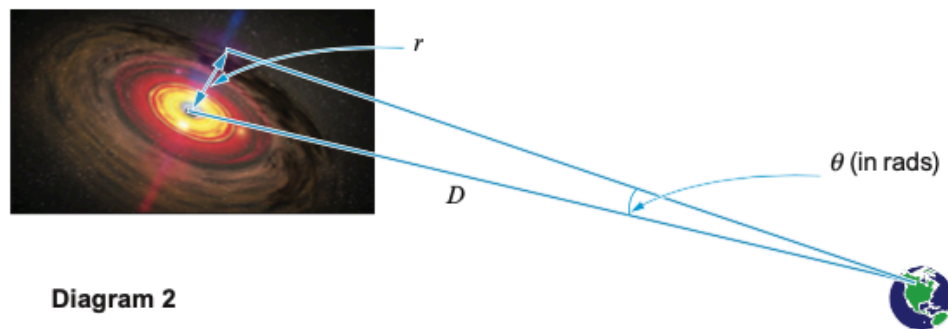


Diagram 2

We can also measure the angular size (θ) of the gas disk (in radians). Given the angular size (θ) and the radius, r , of the gas disk, we can obtain the distance, D by assuming that the angle (θ) is small. The simplicity of this method is remarkable and gives reliable results that are not dependent on controversial or unproven theories. In fact, the results obtained thus far from a couple of suitable galaxies with megamasers leads to a Hubble constant of $(68.9 \pm 7.1) \text{ km s}^{-1} \text{ Mpc}^{-1}$.

7. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) In your own words explain what a megamaser is and how it works. (See paragraphs 1, 4, 7 and 8.) [5]

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(b) If the lifetime of a metastable level leading to visible light is $2\mu\text{s}$, estimate the lifetime of a 22.235 GHz maser transition. (See paragraph 2.) [3]

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- (c) The recessional velocity of the galaxy used in paragraph 10 is $3\,300\text{ km s}^{-1}$. Use Hubble's law to calculate the distance of the galaxy in Mpc. (N.B. use the value of H_0 in paragraph 10. There is no need to change the units.) [2]

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- (d) What does the author mean by “We can also measure the centripetal acceleration, a , of maser clouds by observing how the Doppler shift velocity changes over time”? (See paragraph 9.) [3]

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- (e) (i) Doppler shift measurements are obtained for a gas disk orbiting a black hole. The orbital velocity of gas around the black hole is measured as 410 km s^{-1} and the acceleration of the gas disk is measured as 6 km s^{-1} **per year**. Calculate the distance of this region of the gas disk from the black hole. (See paragraph 9.) [3]

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- (ii) The angular size (θ) in radians of the gas disk is measured as 5.1×10^{-9} rad $\pm 10\%$ and its distance from Earth is measured as 1.53×10^{23} m $\pm 10\%$. Evaluate whether or not these values are consistent with your calculation in part (e)(i). (See Diagram 2.) [4]

Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
7	(a)	Microwave laser or amplifier or equivalent (1) from water [molecules] or water clouds or steam (1) in gas disk around (supermassive) black hole (or quasar) (1) pumping or excitation provided by collisions (accept light) (1) (em radiation) propagates outwards (away from black hole) or reference to population inversion (1) Treat as neutral reference to watermaser. (Don't award the individual marks if later contradicted)	1 1 1	1 1		5		
	(b)	Appropriate wavelength or energy or frequency chosen e.g. 400 nm-700 nm or 2-3 eV or $4-8 \times 10^{14}$ Hz (1) Valid method for obtaining ratio of frequencies, wavelength or energy (1) Answer 35 m[s] – 80 m[s] (1)	1	1 1		3	2	
	(c)	$\frac{3\,300}{68.9}$ (1) 48 [Mpc] (1)		2		2	2	
	(d)	Measure the velocity using Doppler shift or use of Doppler equation (1) at different times (1) acceleration = rate of change of velocity (1) Alternative: Measure the velocity using Doppler shift (1) Obtain $v_{\max}$ (1) $a = \frac{v_{\max}^2}{r}$ or equivalent e.g. use v and r to calculate a (1)			3	3		

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(e)	(i)	Use of $r = \frac{v^2}{a}$ (1) Acceleration conversion i.e. /365/24/3600 (1) Answer = 8.8×10^{14} [m] or 8.8×10^{11} k[m] (1)			3	3	3	
		(ii)	Approximation used i.e. $D = \frac{r}{\theta}$ (1) Answer = 7.8×10^{14} [m] (1) Hence consistent (since overlap) i.e. valid conclusion based on calculations (1) Comparing e.g. 1.53 ± 0.15 and 1.73 ± 0.17 but also accept combined error = 20% or 8.8×10^{14} is less than 20% bigger than 7.8×10^{14} (1)			4	4	2	
			Question 7 total	4	6	10	20	9	0

2018

Read through the following article carefully.

Freely adapted from:

The Physics of Optical Fibres
By Justino Luis Moreno

Paragraph

Basically, an optical fibre is just a piece of glass along which you send light. A similar procedure was first demonstrated not with glass but with water flowing from a spout in 1840 by scientists Daniel Colladon and Jacques Babinet. Babinet later published his work in an article entitled "On the reflections of a ray of light inside a parabolic liquid stream". This effect can be reproduced quite easily in a school lab using the apparatus shown in Figure 1.

1

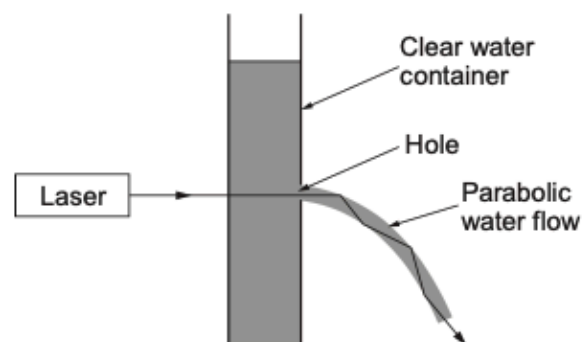


Figure 1

Although this set up is pleasing to the eye and is the basis of some water features, it isn't much use for international telecommunication! A standard optical fibre is shown in Figure 2 along with a ray of light entering it.

2

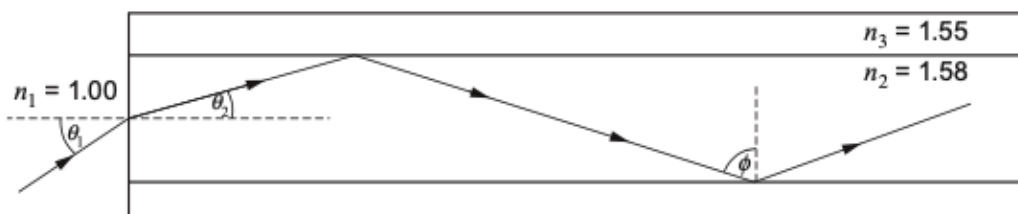


Figure 2

This ray of light is repeatedly reflected along the length of the optical fibre. If the entrance angle (θ_1) is small enough then an effect called total internal reflection (TIR) means that the light is completely reflected each time resulting in no light escaping as it travels along the optical fibre. The physicists in charge of designing these things can prove that the minimum value angle ϕ can have for TIR to take place is given by the equation:

3

$$\phi = \sin^{-1}\left(\frac{n_3}{n_2}\right)$$

This angle is around 80° for the optical fibre shown. The minimum angle for ϕ means that there is a maximum angle for θ_1 . This gives an acceptance cone where the input light gets propagated without loss. A quick bit of geometry shows you that the exit angle is the same as the entry angle so that when light exits it produces an almost perfect cone shaped beam with a circular cross-section as can be seen in Figure 3.



Figure 3

An important term in optical fibre technology is maximum bit rate. Since each pulse represents a bit of data, this is the highest frequency of pulses that can be sent down an optical fibre before the pulses start to overlap and become indistinguishable from each other. A monomode optical fibre cable of length 80 km can comfortably have a maximum bit rate of 10 Gbs^{-1} . This means that 10^{10} pulses can pass along its length every second without overlapping. A typical telephone conversation requires around 10 kbs^{-1} meaning that one monomode optical fibre cable can carry a million telephone conversations simultaneously. High definition TV requires a much higher bit rate and a 10 Gbs^{-1} fibre will only handle around 2 000 high definition TV signals.

One of the most important factors that limit data transfer in optical fibres is multimode dispersion. This, put simply, is to do with the entrance cone of light in Figure 3. There is a range of distances that the pulses have to travel because there are a variety of angles at which they can travel along the fibre. The pulses then become spread out and indistinct, ruining the digital signal. Multimode dispersion is eliminated by using monomode optical fibre cables which have very thin cores. As a rule, monomode fibres have a core diameter of around $8\text{ }\mu\text{m}$. This means that the core is less than 10 wavelengths thick so that the light stops behaving like rays. For monomode fibres there is only one propagation direction – along the axis.

Another drawback of sending signals down long lengths of optical fibres is that some of the light is either scattered or absorbed by the glass molecules themselves (an effect known as attenuation). Although no light escapes the fibre due to TIR there are other losses involved and these are usually summarised by using a decibel (dB) scale. This scale is defined by saying that a 10 dB decrease (-10 dB) in power is when the power has dropped to 10% of its input value. A loss of -20 dB then corresponds to a drop to 1% power and -30 dB is a drop to 0.1% power. The following table shows the relationship between dB values and power ratio:

dB	Power ratio $\left(\frac{P}{P_0}\right)$
-5	0.316
-10	0.100
-15	0.032
-20	0.010

Table 1

The losses of optical fibres are usually quoted in the unit dB/km and some modern optical fibres can have values as low as 0.01 dB/km. Hence, each km of cable loses 0.01 dB meaning that you can use 1 000 km before your signal is down to 10% strength. Optical fibres have come a long way since they were born in a fountain of light nearly 200 years ago. They stretch out (literally) to all areas of the world bringing light, sound and broadband wherever they go. Technical advances mean that data can be sent at a rate of 1.05×10^{15} pulses per second over a distance of 50 km with only one monomode optical fibre. Nonetheless, the technology has its limitations of which attenuation and multimode dispersion are but two.

8. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) Explain why the water in Figure 1 flows in a downward curve (paragraph 1). [2]

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(b) Show that the minimum angle (ϕ) for total internal reflection in the optical fibre of Figure 2 is less than 80° (paragraph 3). [2]

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(c) Explain why the light emerging from the optical fibre is in the shape of a cone of light (paragraphs 3 & 4 and Figures 2 & 3). [2]

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(d) Calculate the angle (α) of the cone of light emerging from the optical fibre in Figure 3 (see paragraphs 3 & 4 and Figures 2 & 3). [4]

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(e) Rhys makes the following claim:

"A multimode optical fibre that can transfer data at a maximum bit rate of 500 Mb s^{-1} over a distance of 1 km would be able to transfer data at a maximum rate of 50 Mb s^{-1} over a distance of 10 km."

Discuss whether or not Rhys's claim is valid (paragraph 5 & 6).

[3]

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(f) A certain type of optical fibre has an attenuation of 0.8 dB/km. When the signal decreases to 6% of its original intensity it must be amplified or the signal will be lost. An engineer intends to install these optical fibres in lengths of 20 km before each amplifier. Determine whether or not this is an appropriate length of optical fibre to use (see paragraph 7 and Table 1). [3]

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(g) Aled claims that a core diameter of around $13 \mu\text{m}$ is thin enough for a monomode optical fibre because a wavelength in air of $1.3 \mu\text{m}$ is usually used in communications. Rhian claims that this is nonsense because the wavelength in the optical fibre is changed due to its refractive index. Justify by including a calculation who is correct (paragraph 6). [4]

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Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
8	(a)	[Constant] velocity horizontal (1) Accelerating / force downwards [b.o.d. motion under gravity] (1) Projectile motion → 1 mark		2		2		
	(b)	$\phi = \sin^{-1}\left(\frac{1.55}{1.58}\right)$ (1) 78.82°, 78.8° or 79° etc. seen (not just 80) (1) 1.38 [rad] → 1st mark	1	1		2	2	
	(c)	Maximum angle for propagation with TIR (1) Rotational symmetry (about central axis) (1)		2		2		
	(d)	$\theta_2 = 90 - 80 = 10$ degree or similar (11.2 degree etc.) (1) $n_1 \sin \theta_1 = n_2 \sin \theta_2$ used (1) Rearrangement $\theta_1 = \sin^{-1}\left(\frac{n_2 \sin \theta_2}{n_1}\right)$ (1) Answer = 17.9° or 15.9° if 10° used (1)	1	1 1 1		4	3	
	(e)	Dispersion increases with length of fibre (1) [Maximum] bit rate and distance proportional [however stated, eg. Accept distance × 10 → bit-rate down by factor of 10] (1) Conclusion consistent with argument (1)			3	3		
	(f)	20 × 0.8 dB or 16 dB seen (1) Correct comparison with table e.g. 15 dB too much (0.03) (1) Correct conclusion: no, signal drops too much / distance too large (1) [NB Either distance for 6% = 15.3 km or 20 km → 2.5 % gives the first two marks].			3	3	2	
	(g)	Wavelength is decreased by factor n or $v=c/n$ (1) Wavelength is 820 nm (1) So thickness of around 8.2 μm is required (1) 13 μm student is wrong (1) [Allow Aled correct because 1.3 μm × 10 = 13 μm for 1 mark]			4	4	2	
		Question 8 total	2	8	10	20	9	0

2019

SECTION B

Answer all questions

Read the following article carefully.

A Brief History of Particle Physics

Freely adapted from <http://www.particleadventure.org/other/history/quantum.html>

Paragraph

The story starts a long time ago (~400BC) in ancient Greece. Democritus, Leucippus and Epicurus developed the theory that *the universe consists of empty space and an (almost) infinite number of invisible particles* which differ from each other in *form, position, and arrangement*. All matter is made of indivisible particles called atoms.

1

It could be argued that very little happened for another 2000 years until in 1654 Otto von Guericke invented a vacuum pump. You will probably have heard about one use he made of this vacuum pump – the Magdeburg hemispheres. Two metal hemispheres with a vacuum between them could not be separated by teams of horses pulling them apart.

2



Otto von Guericke also invented an electrostatic generator – electron accelerators were born and lightning discharge could be produced artificially.

3

In 1705, it was noted that by combining both inventions of Otto von Guericke, lightning discharge can go between electrodes in a gas and can go further at low pressures.

4

In 1838, Michael Faraday noted a strange glow from low pressure gases in glass tubes when the gas was conducting electricity. Some 40 years later Eugen Goldstein noted so-called "cathode rays" in these gases. In 1898, Joseph (JJ) Thomson identified these cathode rays as streams of negatively charged particles and made measurements on the properties of these electrons. He then put forth his "plum-pudding" model of the atom. In this model, the atom is a slightly positive sphere with small, raisin-like negative electrons inside – he obviously didn't know that radioactivity, researched by Marie Curie, was a nuclear phenomenon. Hardly surprising since the nucleus hadn't been discovered yet....

5

The early 20th century saw Hans Geiger and Ernest Marsden, under the supervision of Ernest Rutherford, scatter 4.7 MeV alpha particles off a gold foil and observe large angles of scattering. This put an end to the plum-pudding model and suggested that atoms have a small, dense, positively charged nucleus of diameter around 10^{-14} m surrounded by electrons orbiting at a distance of around 10^{-10} m.

6

7. Answer the following questions in your own words. Extended quotes from the original article will not be awarded marks.

- (a) Calculate the maximum force required to separate two Magdeburg hemispheres of radius 10 cm (atmospheric pressure = 1.01×10^5 Pa, see paragraph 2). [2]

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- (b) Suggest why lightning discharge travels further at low pressures (see paragraphs 4 & 5). [2]

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- (c) (i) Calculate the distance of nearest approach for a 4.7 MeV alpha particle when fired at gold foil (the charge on a gold nucleus is $+79e$, see paragraph 6). [3]

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- (ii) State why your answer to (c)(i) and the results of the alpha particle scattering experiment contradict the "plum-pudding" model (see paragraphs 5 & 6). [2]

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- (d) A student claims that a 4 keV electron is suitable for atomic diffraction experiments. Determine whether or not she is correct (see paragraphs 6 & 8). [4]

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- (e) Explain why the strong nuclear force is required to hold nuclei together (see paragraphs 6 & 9). [1]

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- (f) A typical thermal electron has a rms speed of approximately $100\,000\text{ ms}^{-1}$. When a positron annihilates an electron at room temperature, two photons of wavelength $2.43 \times 10^{-12}\text{ m}$ are produced. Explain, using conservation of momentum, why these photons travel in opposite directions (see paragraph 10). [3]

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(g) Explain why the quark make up of a π^- meson is $\bar{u}d$. [1]

(h) Calculate the mass of a top quark in kg (see paragraph 15).

Note that $1 \text{ u} \equiv 931 \text{ MeV}$ if required. [2]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Use of $F = Ap$ and $A = \pi r^2$ or accept $A = 4\pi r^2$ (1) Correct answer = 3 173 [N] (1) [no ecf from use of $A = 4\pi r^2$]	1	1		2	2	
	(b)		Fewer collisions (1) ...because greater distances between molecules (or smaller density or more free space) (1)		2		2		
	(c)	(i)	Application of conservation of energy i.e. $E_k = \frac{Qq}{4\pi\epsilon_0 r}$ (1) Conversion of 4.7 MeV $\rightarrow$ J i.e. $4.7 \times 10^6 \times 1.6 \times 10^{-19} = 7.52 \times 10^{-13} \text{ J}$ (1) Answer = $4.8 \times 10^{-14} \text{ [m]}$ (1)		3		3	3	
		(ii)	Smaller than atomic radius or inside plum pudding (1) So force / PE never great enough (for rebound) or scattering angle too large in experiment (1)			2	2		
	(d)		Use of conservation of energy to get speed or momentum e.g. $p^2 = 2mE_k$ etc. $v = 3.75 \times 10^7 \text{ [m s}^{-1}\text{]}$ or $p = 3.41 \times 10^{-22} \text{ [Ns]}$ (1) Calculation of a wavelength using $\lambda = \frac{h}{p}$ (even if incorrect, $1.94 \times 10^{-11} \text{ m}$ is the correct value) (1) Comparison of the calculated wavelength with atomic separation (or 10^{-9} to 10^{-11} m) (1) Correct final conclusion and correct wavelength ($1.94 \times 10^{-11} \text{ m}$) (1)			4	4	3	
	(e)		Proton repulsion or like charges repel etc.	1			1		
	(f)		Photon mom calculated $\left(p = \frac{h}{\lambda}\right) = 2.73 \times 10^{-22} \text{ [kg m s}^{-1}\text{]}$ (1) Electron momentum calculated = $9.11 \times 10^{-26} \text{ [kg m s}^{-1}\text{]}$ (1) [Initial momentum negligible] so final momenta must cancel (1)			3	3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(g)		Charge of $\bar{u}d = -\frac{2}{3} - \frac{1}{3}$	1			1		
	(h)		Either: $\text{Mass} = \frac{172 \text{ GeV}}{931 \text{ MeV u}^{-1}} = 185 \text{ [u]} \text{ (1)}$ $\therefore \text{Mass} = 185 \text{ [u]} \times 1.66 \times 10^{-27} \text{ [kg u}^{-1}] = 3.07 \times 10^{-25} \text{ [kg]} \text{ (1)}$ Or: $\text{Mass energy} = 172 \text{ GeV} \times 1.60 \times 10^{-19} \text{ J eV}^{-1} = 2.75 \times 10^{-8} \text{ [J]} \text{ (1)}$ $\therefore \text{Mass} = \frac{2.75 \times 10^{-8} \text{ [J]}}{(3.00 \times 10^8 \text{ [m s}^{-1}])^2} = 3.06 \times 10^{-25} \text{ [kg]} \text{ (1)}$		2		2	2	
			Question 7 total	3	8	9	20	12	0

2022

6. Read through the following article carefully.

How do you produce really low temperatures and can you then do some cool stuff with it?

Paragraph

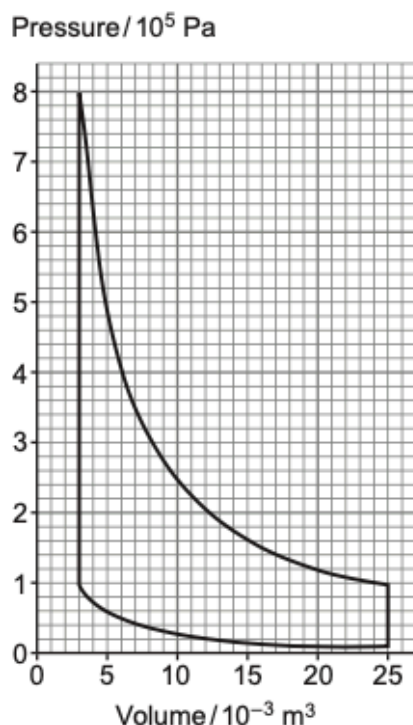


Figure 1

The first thing you have to do is to produce low temperatures and this is usually achieved by taking a gas through a closed cycle such as that shown in Figure 1. 1

1. Air, initially under high pressure, is expanded at constant temperature. 1
2. Heat is allowed to escape at constant volume. 2
3. The gas is compressed at a constant low temperature.
4. The gas is heated at constant volume to a high pressure.

The end result is that the gas does a large amount of work in each cycle. This means that the gas must have heat flowing into it leading to the cooling of whatever it is that requires cooling. 3

A different and simpler method of cooling gases is called the Joule-Thompson method, which does not involve a gas going through a cycle. A gas is compressed to a pressure of two thousand times atmospheric pressure. The gas is then allowed to cool back to room temperature. After that, it is allowed to expand quickly through a nozzle to atmospheric pressure leading to a tremendous amount of further cooling. In the expansion, the gas does a very large amount of work but the non-ideal gas molecules also have an enormous increase in potential energy because of the attractive forces between molecules. Both these factors cool the gas by a huge factor. 4

Okay, so now we have the basic physics for producing low temperatures. Apply this low temperature to air and we can make a product that can give hours of fun – liquid nitrogen (LN_2). 5

Most of the fun that can be obtained from LN_2 is based on the fact that it boils at a temperature of -195.8°C and, in doing so, expands by a factor of approximately 1000. For example, a cool rotating rocket can be built using a simple plastic water bottle. 6

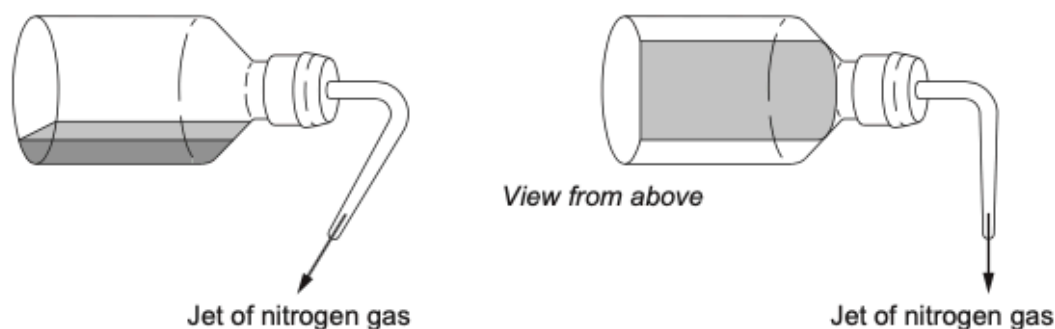


Figure 2

Paragraph

The boiling nitrogen makes the nozzle of the plastic water bottle into a jet engine. When you then place the plastic water bottle on a nicely polished wooden floor it will rotate far more quickly than you would ever anticipate, providing a great demonstration for even the most ardent physics-hating pupil. What can be even more spectacular is when the bottle accidentally hits something solid (like the leg of a lab desk). The wall of the plastic water bottle will be at around -200°C and fragile. This, combined with the high pressure inside the bottle, leads to rather a loud and impressive bang. 7

Although LN_2 can be fun, it can also be dangerous. For instance, when transporting a large, open container of LN_2 (see Figure 3), you should not remain in the confined space of a lift with the container in case the lift becomes stuck. Most of the danger from LN_2 arises from its 200°C temperature difference from its surroundings. Nonetheless, it is always amusing when TV chefs wear safety goggles and gloves when making ice cream using LN_2 but never bother with any safety equipment when using litres of boiling fat above a gas fire. I promise you that the boiling fat is far more dangerous even though the temperature difference, compared with human body temperature, is around 200°C for both LN_2 and boiling fat. 8

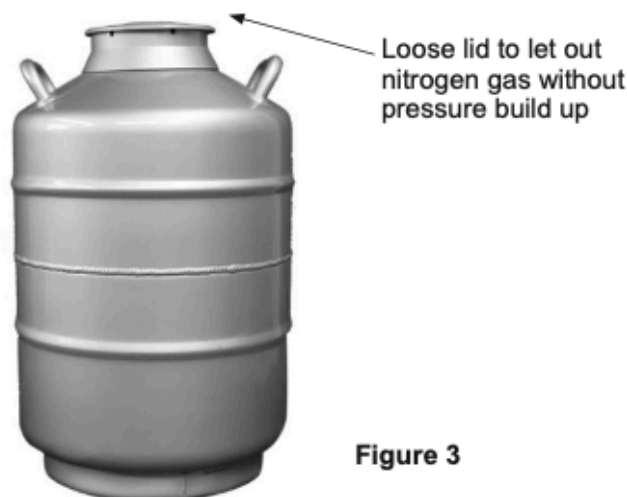
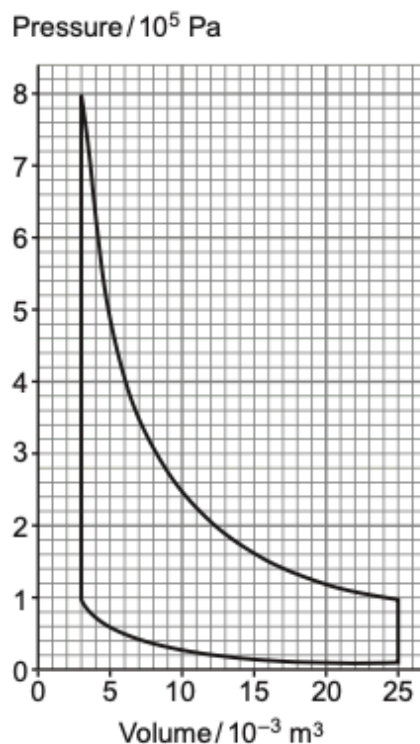


Figure 3

In conclusion, low temperatures and LN_2 can be dangerous for some obvious reasons but they can also lead to some amusement and even some Michelin-starred ice cream. 9

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

- (a) **Label the diagram** with the processes 1–4 along with an arrow to show the direction of the cycle (see paragraph 2). [2]



- (b) The author states that work is done by the gas in the cycle. By considering the four stages of the cycle, explain whether or not this statement is correct (see paragraph 3 and Figure 1). [2]

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(c) Estimate the work done during the cycle. [3]

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(d) When a gas is compressed in a metal cylinder it will become hot. Explain why the gas will then cool to room temperature (see paragraph 4). [2]

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(e) Explain whether or not the author is correct to state that the gas expanding quickly through the nozzle will cool greatly (see paragraph 4). [4]

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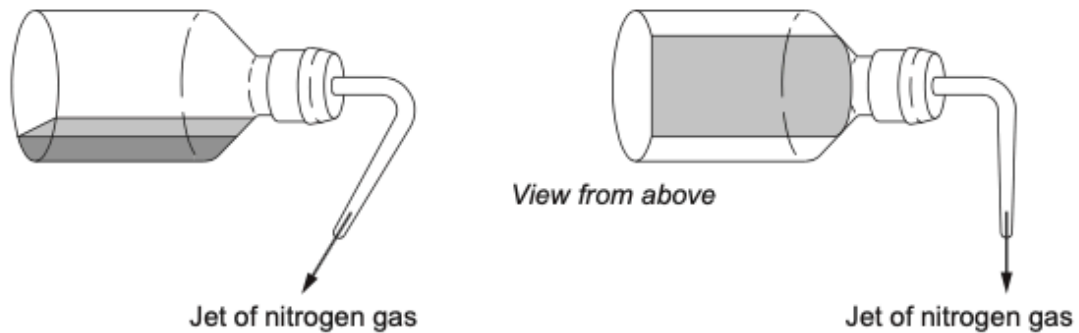
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(f) Explain very briefly why nitrogen gas escapes through the nozzle of the plastic water bottle (see Figure 2 and paragraph 7). [1]

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- (g) Starting from Newton's 3rd Law, explain why the plastic water bottle shown will rotate anti-clockwise. [2]



- (h) An exploding bottle produces a "loud bang" (see paragraph 7). Explain briefly why this "loud bang" can be heard more than 100 m from the exploding bottle. [2]

- (i) Justify the author's statement that it would be dangerous to share a confined space (like a lift) with a large container of liquid nitrogen (see paragraph 8 and Figure 3). [2]

END OF PAPER

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		1 clockwise arrow (1) 1, 2, 3, 4 in correct positions 1 (1) 4 2 3		2		2		
	(b)		True because clockwise OR process 1 has greatest work than 3 (1) No work done for 2 & 4 (1) Apply ecf if needed			2	2		
	(c)		Attempt made at any area under the graph or enclosed or counting squares (1) 9.25ish large squares or 9×25 small squares or correct trapezium / enclosed area method (1) Answer = $4\,600 \pm 400$ [J] (1)			3	3	3	
	(d)		Heat flows from gas to surroundings (through piston) OR conduction through metal (1) Accept molecules collide with container [Thermal] equilibrium or same temperature as surroundings (1)		1	1	2		
	(e)		Any 4 × (1) from: - Volume increases [for expansion part] - Work done by gas [for expansion but not compression] - Internal energy decrease [for expansion but not compression] - Increased PE linked to lower temperature or expansion - KE or internal energy decrease linked to cooling or lower temperature or expansion [Process] too quick for heat transfer			4	4		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(f)		Nitrogen boils OR heat transferred to LN ₂ OR high pressure inside bottle. Accept gas expands	1			1		
	(g)		Force on bottle / nozzle is equal and opposite to force on gas / nitrogen (1) Provides [anticlockwise] moment / torque [for rotation] (1) For 2 nd mark, do not accept "principle of moments" because this is relevant to equilibrium or adding moments. Don't accept equal and opposite forces acting on the walls	1	1		2		
	(h)		Sound [waves] produced (or implied) (1) Implication of pressure release / molecule or gas release / energy release OR displacement of particles/molecules OR vibration of particles / molecules (1)	1	1		2		
	(i)		Nitrogen escapes from the container OR nitrogen escapes into lift (1) ACCEPT pressure in lift builds up DON'T ACCEPT container explodes OR container acts like rocket Leading to suffocation / anything to do with breathing N ₂ (1)			2	2		
			Question 6 total	3	5	12	20	3	0

7. Read through the following article carefully.

Paragraph

A summary of binary stars

First of all, we call two stars orbiting their common centre of mass a “binary star”. Binary stars fall into 5 different classes, categorised by the detection method: 1

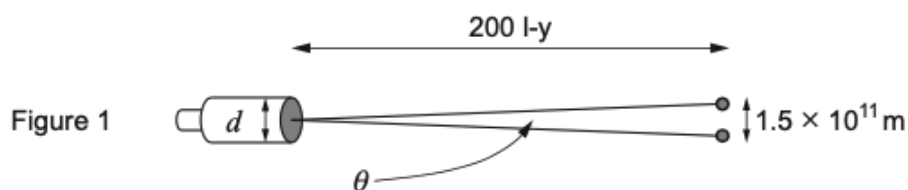
1. Visual binaries
2. Spectroscopic binaries
3. Eclipsing binaries
4. Astrometric binaries
5. Non-eclipsing binaries that can be detected using photometry.

Visual binaries

These do what it says on the tin – the angular separation of the stars is great enough that both stars can be seen as separate objects through a telescope. There is a simple relationship between the minimum angular separation θ (in radians), the diameter, d , of the telescope and the wavelength, λ , of the radiation used. 2

$$\theta = 1.22 \frac{\lambda}{d} \quad \text{Equation 1}$$

This equation means that a nearby binary star that is around 200 light years (l-y) away will only be visible as two stars with a $d = 8.0\text{ m}$ telescope if the two stars are separated by at least the Earth-Sun distance of $1.5 \times 10^{11}\text{ m}$ (see Figure 1). 3



Perhaps the most famous of visual binary stars is the brightest star in the night sky – Sirius. Sirius is not, in fact, a single star but a binary consisting of Sirius A and Sirius B that orbit each other with a period of around 50 years. Sirius A is a main sequence star which has roughly 2 times the mass of the Sun, 1.7 times the Sun's temperature and 25 times its luminosity. Sirius B is a white dwarf with a similar mass to our Sun, a surface temperature 4.3 times greater than our Sun but a luminosity 18 times less than our Sun. 4



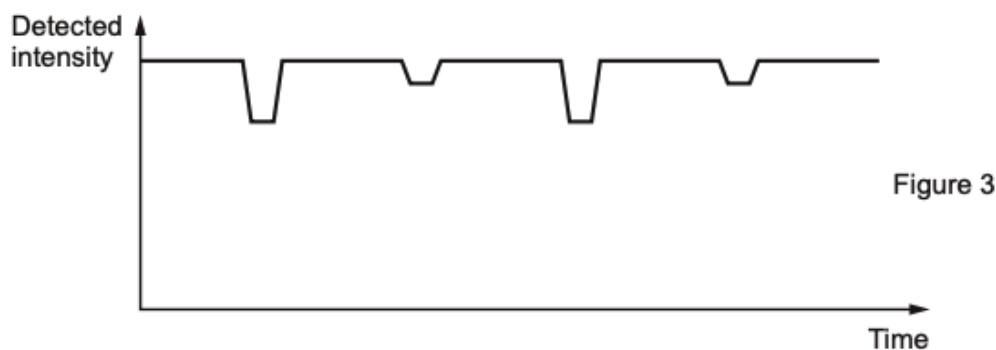
Figure 2

Spectroscopic binaries

These are binaries where the stars cannot be viewed separately but the existence of the two stars can be inferred from the variation of the Doppler shifted light from the system. Both stars will orbit their mutual centre of mass and hence their Doppler shifts will vary periodically during their orbits. This theory is the same as that used to discover exoplanets from the wobble of their parent star. 5

Eclipsing binaries

These happen when the line of sight of the observer is in the plane of the orbit of the two stars. When this happens, one star can pass directly in front of the other leading to a drop in the intensity of light entering the observer's telescope. An idealised graph of intensity against time is shown in Figure 3. 6



In general, there are two eclipses for each orbit. The larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star. 7

Astrometric binaries

These are binary systems where only one star is visible and that star can be seen moving in an orbit around something which is invisible. These are different from spectroscopic binaries because the visible star can actually be seen to move rather than inferring its motion from the variation of the Doppler shift. Stars orbiting neutron stars have been observed by this method but the ultimate goal of this technique would be to observe a star orbiting a black hole. 8

Non-eclipsing binaries that can be detected using photometry

This simply means detecting binaries by measuring light intensity but not via the eclipsing method described earlier. It turns out that there are three different techniques of finding binaries within this class of binary. First, an increased light intensity can be detected periodically from the binary system – this is caused by light being reflected from one star, off the other and into the Earth's telescope. Second, one star can exert such a strong gravitational pull on the other that it becomes deformed – this leads to a periodic fluctuation in the light intensity detected. Third, is an effect sometimes known as Doppler beaming – a star moving towards an observer will lead to an increased number of photons detected per second (as well as a decrease in the wavelength). 9

These 5 different classes of binary stars can provide an enormous amount of valuable data regarding stellar orbits and masses but they can also allow measurement of galactic distances and even give insights into black holes and supernovae. 10

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

- (a) A light year is 9.46×10^{15} m. Use equation 1 and a wavelength of 500 nm to show that the author's claim about an 8 m telescope viewing the binary star mentioned in paragraph 3 and Figure 1 is correct.

[4]

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- (b) (i) Show that the distance between Sirius A and Sirius B is approximately 3×10^{12} m (see paragraph 4). [Mass of the Sun is 2×10^{30} kg.]

[2]

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- (ii) Show that the centre of mass of the Sirius binary system is located approximately 1×10^{12} m from Sirius A.

[2]

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- (iii) Use data in paragraph 4 and Stefan's law to calculate the ratio of the radii of Sirius A and Sirius B: [4]

$$\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}}$$

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- (c) (i) Evaluate whether the variation of light intensity shown in Figure 3 is consistent with an eclipsing binary system (see paragraph 6). [3]

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- (ii) Use Stefan's law to explain why the "larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star" (see paragraphs 6 and 7 and Figure 3). [2]

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- (d) Explain why a star orbiting a much more massive black hole is an astrometric binary (see paragraph 8). [1]

- (e) One of the three methods for detecting non-eclipsing binaries that can be detected by photometry, suggests that stars do not behave as ideal black bodies. Explain which **one** of the three methods relies on stars not behaving as black bodies (see paragraph 9). [2]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Converting ly to m i.e. $200 \times 9.46 \times 10^{15} = 1.892 \times 10^{18}$ [m] (1) Calculating θ in radian i.e. $\frac{1.5 \times 10^{11}}{1.892 \times 10^{18}} = 7.93 \times 10^{-8}$ (1) (can be done by using tan or trig) $\theta = 1.22 \frac{\lambda}{d}$ used (whatever the subject) i.e. 7.6 m for visible light or 7.625×10^{-8} seen (1) Final calculation leading to suitable conclusion (1) Award full marks for a calculation of 1.45×10^{11} seen and compared with 1 AU	1	1 1 1		4	3	
	(b)	(i)	Substitution into period equation e.g. even $50 = 2\pi \sqrt{\frac{d^3}{G \times 3 \times 2 \times 10^{30}}}$ (1) Final rearrangement and correct substitution seen or 2.93×10^{12} [m] (1)	1	1		2	1	
		(ii)	Substitution into C of M equation (1) can be implied $\frac{1}{3} \times 2.93 \times 10^{12} = 0.977 \times 10^{12}$ or equivalent seen (1)	1	1		2	1	
		(iii)	Use of Stefan's law (1) Use of area proportional to R^2 (1) $\frac{R_A^2 T_A^4}{R_B^2 T_B^4} = \frac{R_A^2 1.7^4}{R_B^2 4.3^4} = \frac{25}{1/18}$ (1) Answer = $\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}} = 136$ or 140 (1)	1	1 1 1		4	3	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	Constant intensity when both stars seen or gradual drop or partial eclipse or regular time intervals (1) One star in front of the other, light is "blocked" (1) Different dips depending on which star is "blocked" (1)			3	3		
		(ii)	{Intensity / power / luminosity / brightness} increases with T (1) The same area is blocked regardless of which star is doing the blocking or $I = \frac{P}{A} = \sigma T^4$ (1)			2	2		
	(d)		Black hole is not visible [and star orbiting is]			1	1		
	(e)		Reflection method (1) accept the first method of the 3 Black bodies do not reflect OR black bodies absorb all em radiation incident (1) Award one mark only for existence of absorption lines in the spectrum			2	2		
			Question 7 total	4	8	8	20	8	0